Categorical Homotopy Theory - Self Study

Grant Talbert

Summer, 2025

BOSTON UNIVERSITY

Abstract

This text contains my notes on Categorical Homotopy Theory, by Emily Riehl. I read this book independently during the summer of 2025, between my sophomore and junior years. My notes largely paraphrase the text. Going into the text, I lacked a precise understanding of model category theory, and had a weak intuition for simplicial sets/objects and methods. I also lacked some helpful algebraic topology knowledge. I continuously supplimented my readings with references to Hatcher's *Algebraic Topology*, Hovey's *Model Category Theory*, and on occasion May's *Simplicial Objects in Algebraic Topology*. I intended to, as a side effect of reading this text, improve my understanding of these subjects such that I might only need to skim texts on them in the future, rather than read them in their entirety.

Notation. Disjoint unions are denoted by $\amalg$. Composition of functors and morphisms is denoted by $\circ$, or juxtaposition when the notation can be is well understood. Vertical composition of 2-cells is denoted by $\circ$, and horizontal composition is denoted by $\cdot$. Usually, juxtaposition of 2-cells denotes horizontal composition, but on occasion it may denote vertical composition if it can be well understood that there is no horizontal composition. For a category $\mathcal{C}$, I also write $\mathcal{C}$ to denote the set of objects of $\mathcal{C}$. On occasion, I will instead write $\mathrm{Obj}(\mathcal{C})$ for emphasis. I write $\mathrm{Mor}(\mathcal{C})$ for the set of arrows of $\mathcal{C}$.

Contents

I Derived functors and homotopy (co)limits **2**

1 All concepts are Kan extensions **3**
- 1.1 Kan Extensions . . . 3
- 1.2 A formula . . . 4
- 1.3 Pointwise Kan extensions . . . 4
- 1.4 All concepts . . . 4
- 1.5 Adjunctions involving simplicial sets . . . 5

2 Derived functors via deforamtions **7**
- 2.1 Homotopical categories and derived functors . . . 7
- 2.2 Derived functors via deformations . . . 10
- 2.3 Classical derived functors between abelian categories . . . 12
- 2.4 Preview of homotopy limits and colimits . . . 13

3 Basic concepts of enriched category theory **14**
- 3.1 A first example . . . 14
- 3.2 The base for enrichment . . . 14
- 3.3 Enriched categories . . . 15
- 3.4 Underlying categories of enriched categories . . . 18
- 3.5 Enriched functors and enriched natural transformations . . . 22
- 3.6 Simplicial Categories . . . 34
- 3.7 Tensors and Cotensors . . . 35
- 3.8 Simplicial homotopy and simplicial model categories . . . 40

Part I

Derived functors and homotopy (co)limits

Chapter 1

All concepts are Kan extensions

I already know Kan extensions so I will breeze through this chapter.

1.1 Kan Extensions

Definition 1.1 (Lan Extension)**.** Given functors $F : \mathcal{C} \to \mathcal{E}$ and $K : \mathcal{C} \to \mathcal{D}$, a *left Kan extension* of F along K is a functor $\operatorname{Lan}_K F : \mathcal{D} \to \mathcal{E}$ and a natural transformation $\eta : F \Rightarrow \operatorname{Lan}_K F \circ K$ (a pasting diagram) which is initial with respect to this property; for any other extension $G : \mathcal{D} \to \mathcal{E}$ and $\gamma : F \Rightarrow G \circ K$, there is a unique natural transformation $\nu : \operatorname{Lan}_K F \Rightarrow G$ such that $\gamma \cdot 1_K = (\nu \cdot 1_F) \circ \eta$. Dually, a right Kan extension is terminal with respect to natural transformations $\varepsilon : \operatorname{Ran}_K F \circ K \Rightarrow F$. ♡

Remark. These definitions generalize to any 2-category. We usually specialize in the 2-category $\mathcal{C}\mathrm{at}$ for obvious reasons, so we reserve the term "Kan extension" for $\mathcal{C}\mathrm{at}$. When working in $\mathcal{C}\mathrm{at}$, I use "extension" to mean a pair (G, γ) satisfying the extension property, but not necessarily universally.

In unenriched category theory, universal properties (in this text) are representations of functors. A left Kan extension $\operatorname{Lan}_K F$ with notation as above is a representation for

$$\mathcal{E}^{\mathcal{C}}(F, - \circ K) : \mathcal{E}^{\mathcal{C}} \to \mathcal{S}\mathrm{et}.$$

To justify this, recall here that universality gives for any G,

$$\mathcal{E}^{\mathcal{C}}(F, G \circ K) \cong \mathcal{E}^{\mathcal{D}}(\operatorname{Lan}_K F, G).$$

Hence, $\operatorname{Lan}_K F$ is a representation as the universal object. Alternatively, the Yoneda lemma promotes any extension $\gamma : F \Rightarrow G \circ K$ to a natural transformation

$$\gamma : \mathcal{E}^{\mathcal{C}}(G \circ K, -) \Rightarrow \mathcal{E}^{\mathcal{C}}(F, -).$$

Precomposing by $1 \times K$ gives

$$\gamma \cdot 1 : \mathcal{E}^{\mathcal{C}}(G \circ K, - \circ K) = \mathcal{E}^{\mathcal{D}}(G, -) \Rightarrow \mathcal{E}^{\mathcal{C}}(F, - \circ K).$$

Universality is then equivalent to that the natural transformation

$$\eta : \mathcal{E}^{\mathcal{D}}(\operatorname{Lan}_K F, -) \Rightarrow \mathcal{E}^{\mathcal{C}}(F, - \circ K)$$

be an isomorphism.

By universality, $\operatorname{Lan}_K -$ is functorial provided it always exists, so this gives an adjunction for precomopsition K^*:

$$\mathcal{E}^{\mathcal{D}}(\operatorname{Lan}_K F, G) \cong \mathcal{E}^{\mathcal{C}}(F, G \circ K) \qquad \operatorname{Lan}_K \dashv K^* \dashv \operatorname{Ran}_K.$$

1.2 A formula

Theorem 1.1. *Let $\mathcal{C}$ small, $\mathcal{D}$ locally small, and $\mathcal{E}$ cocomplete. Then any $F : \mathcal{C} \to \mathcal{E}$ has an extension along any $K : \mathcal{C} \to \mathcal{D}$ given pointwise by*

$$\mathrm{Lan}_K F(d) = \int^{c \in \mathcal{C}} \mathcal{D}(K(c), d) \cdot F(c)$$

for $\cdot$ the copower/tensor and $d \in D$. Dually,

$$\mathrm{Ran}_K F(d) = \int_{c \in \mathcal{C}} F(c)^{\mathcal{D}(K(c),d)}.$$

Here, exponentiation is the power/cotensor.

Proof. See my notes for Mac Lane. ■

Theorem 1.2. *With notation as in 1.1, we also have*

$$\mathrm{Lan}_K F(d) = \varinjlim \left(F \circ \Pi^d \right), \qquad \Pi^d : (K \downarrow d) \to \mathcal{C},$$

for Π^d the comma category projection. We will prove these formulas again in the context of weighted limits later. Dually,

$$\mathrm{Ran}_K F(d) = \varprojlim \left((d \downarrow K) \xrightarrow{\Pi_d} \mathcal{C} \xrightarrow{F} \mathcal{E} \right).$$

1.3 Pointwise Kan extensions

Definition 1.2 (Preserves)**.** A functor $L : \mathcal{E} \to \mathcal{F}$ preserves left Kan extensions if $(L\,\mathrm{Lan}_K F, L\eta)$ is the left Kan extension $\mathrm{Lan}_K LF$. ♡

The following holds even when coends and colimits don't exist. Again, see my notes on Mac Lane for the (simple) proof.

Lemma 1.1. *Left adjoints preserve left Kan extensions. Right adjoints preserve right Kan extensions.*

It has generally been decided that Kan extensions in full generality are not nearly as important as the following special case, and the acompanying theorem which reconsiles the terminology.

Definition 1.3 (Pointwise Kan extensions)**.** If $\mathcal{E}$ is locally small, a pointwise right Kan extension is one preserved by all representable functors $\mathcal{E}(e, -)$. By contravariance, a pointwise left Kan extension is one such that postcomposition by $\mathcal{E}(-, e)$ with the opposite (right) Kan extension gives a right Kan extension. ♡

Theorem 1.3. *A right Kan extension is pointwise if and only if it is computed by limits as described in theorem 1.2. A left Kan extension is pointwise if and only if it is computed by colimits as in theorem 1.2.*

1.4 All concepts

The notion of Kan extensions subsumes all the other fundamental concepts of category theory.

Theorem 1.4. *Let $F : \mathcal{C} \to \mathcal{E}$. A limit of F is precisely a right Kan extension of F along the terminal functor $! : \mathcal{C} \to \mathbf{1}$. A left Kan extension along the terminal functor is nprecisely a colimit.*

Theorem 1.5. *Adjunctions $(F, G, \eta, \varepsilon)$ are precisely left Kan extensions (G, η) of 1 along F such that F preserves the Kan extension.*

Kan extensions also give the Yoneda lemma. For $F : \mathcal{C} \to \mathcal{S}\mathrm{et}$,

$$F(c) \cong \mathrm{Ran}_{1_\mathcal{C}} F(c) \cong \int_{x\in\mathcal{C}} F(x)^{\mathcal{C}(c,x)} \cong \int_{x\in\mathcal{C}} \mathcal{S}\mathrm{et}(\mathcal{C}(c,x), F(x)) \cong \mathcal{S}\mathrm{et}^{\mathcal{C}}(\mathcal{C}(c,-), F).$$

The first isomorphism is kind of just obvious, the second is the pointwise construction via ends, the third is definition of exponentials in $\mathcal{S}\mathrm{et}$, and the fourth is an identity I proved in my notes on Mac Lane. This also gives us the coYoneda lemma:

$$F(c) \cong \int^{x\in\mathcal{C}} \mathcal{C}(x,c) \cdot F(x).$$

Finally, if $\mathcal{E}$ is complete and K is fully faithful, then the counit is an isomorphism $\mathrm{Ran}_K F \circ K \cong F$.

1.5 Adjunctions involving simplicial sets

We denote by $\mathrm{s}\mathcal{S}\mathrm{et}$ the functor category $\mathrm{Fun}(\mathbf{\Delta}^{\mathrm{op}}, \mathcal{S}\mathrm{et})$, where $\mathbf{\Delta}$ is the simplex category, and $\mathbf{\Delta}_+$ is the augmented simplex category which includes $[-1]$. $\mathrm{s}\mathcal{S}\mathrm{et}$ is the free colimit completion of $\mathbf{\Delta}$, although this is not unique to $\mathbf{\Delta}$ and holds for any small category.

Let $\mathcal{E}$ be cocomplete and locally small, and let $\Delta^\bullet : \mathbf{\Delta} \to \mathcal{E}$ be covariant. Write Δ^n for the image $\Delta^\bullet[n]$; similarly for $X \in \mathrm{s}\mathcal{S}\mathrm{et}$, write X_n for the set $X[n]$ of *n-simplices* and $X_\bullet : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{S}\mathrm{et}$ when we wish to emphasize it's functorial role, rather than its role as an assignment of sets. Define $L = \mathrm{Lan}_y \mathbf{\Delta}^\bullet$ as the left Kan extension of $\Delta^\bullet$ along the Yoneda embedding y:

$$\begin{tikzcd}
\mathbf{\Delta} \arrow[rr, "\Delta^\bullet"] \arrow[dr, "y"'] & \arrow[d, phantom, "\Downarrow\cong"] & \mathcal{E} \\
& \mathrm{s}\mathcal{S}\mathrm{et} \arrow[ur, dashed, "L"'] &
\end{tikzcd}$$

Since the Yoneda embedding is fully faithful and $\mathcal{E}$ is locally small and cocomplete, the unit is an isomorphism by the dual of the above statement. The functor L is then defined via theorem 1.1 as

$$L(X) := \int^{n\in\mathbf{\Delta}} \mathrm{s}\mathcal{S}\mathrm{et}(y(n), X) \cdot \Delta^n \cong \int^{n\in\mathbf{\Delta}} X_n \cdot \Delta^n.$$

The first isomorpohism is once again theorem 1.1, and the second is the Yoneda lemma since $y(n) = \mathbf{\Delta}(-, n)$ and indeed $\mathrm{Nat}(\mathbf{\Delta}(-,n), X) \cong X(n) = X_n$. Since coends are colimits, which are characterized by coproducts and coequalizers, we obtain an explicit description as

$$L(X) = \mathrm{coeq}\left(\coprod_{f\in\mathrm{Mor}(\mathbf{\Delta})} X_{\mathrm{cod}\, f} \cdot \Delta^{\mathrm{dom}\, f} \rightrightarrows \coprod_{[n]\in\mathbf{\Delta}} X_n \cdot \Delta^n \right).$$

The definition of L on the arrows is given by universality. We usually choose to denote by Δ^n the simplicial set $y[n]$, and by cocompleteness we have

$$(L \circ y)[n] = \Delta^n,$$

somewhat justifying this different notation we have used here.

Since colimits commute, L preserves colimits, so the adjoint functor theorem gives an adjoint $R : \mathcal{E} \to \mathrm{s}\mathcal{S}\mathrm{et}$, which has for any $e \in \mathcal{E}$,

$$R(e)_n \cong \mathrm{s}\mathcal{S}\mathrm{et}(y[n], R(e)) \cong \mathcal{E}(Ly[n], e) \cong \mathcal{E}(\Delta^n, e).$$

The first isomorphism is Yoneda, the second is the adjunction relation, and the third is our above discussion. Thus, n-simplices of $R(e)$ can be defined as the set of maps from $\Delta^n \to e$ in $\mathcal{E}$, making $R = \mathcal{E}(\Delta^\bullet, -)$ with a mild abuse of notation. Face and degeneracy maps for the simplicial set $R(e)$ are also thus given by precomposition by the appropriate maps to Δ^n (called a cosimplicial object since it is covariant). Levelwise postcomposition makes R functorial.

The geometric realization of general simplicial sets is as follows. There is a natural functor $\mathbf{\Delta} \to \mathcal{T}\mathrm{op}$ given by $[n] \mapsto \Delta^n$ for Δ^n the standard n-simplex. It has both left- and right-adjoints. Since $\mathcal{T}\mathrm{op}$ is cocomplete and locally small, plugging in $\mathcal{T}\mathrm{op}$ for $\mathcal{E}$ in the previous discussion gives an adjunction $L \dashv R$ with $L : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{T}\mathrm{op}$ the left Kan extension, and $R : \mathcal{T}\mathrm{op} \to \mathrm{s}\mathcal{S}\mathrm{et}$ it's right adjoint. R is called the *total singular complex functor*, denoted usually by S, and L is the *geometric realization*, denoted usually by $|-|$. It's computed as

$$|X| = \int^{n \in \mathbf{\Delta}} X_n \times \Delta^n = \mathrm{coeq}\left(\coprod_{f \in \mathrm{Mor}(\mathbf{\Delta})} X_{\mathrm{cod}\, f} \cdot \Delta^{\mathrm{dom}\, f} \rightrightarrows \coprod_{[n] \in \mathbf{\Delta}} X_n \cdot \Delta^n \right).$$

Here we note that copower is isomorphic to cartesian proeuct in $\mathcal{T}\mathrm{op}$. This gives $|y(n) = \Delta^n| = \Delta^n$, once again justifying the annoying notation.

Let $\Delta^\bullet : \mathbf{\Delta} \to \mathcal{C}\mathrm{at}$ now be the functor mapping $[n]$ to the poset category $n+1$ and maps to functors. Indeed, $\Delta^\bullet$ is a full embedding $\mathbf{\Delta} \hookrightarrow \mathcal{C}\mathrm{at}$. It's right adjoint this time is the *nerve* functor $N : \mathcal{C}\mathrm{at} \to \mathrm{s}\mathcal{S}\mathrm{et}$. Recall that $N(\mathcal{C})$ is a simplicial set with an n-simplex given by a string of n composable arrows in $\mathcal{C}$, or equivalently any functor $[n] \to \mathcal{C}$. The th degeneracy maps insert the identity arrow, and the ith face map composes i and $i+1$, unless $i = 0$ or n, in which case the arrow is dropped.

The left adjoint $h : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{C}\mathrm{at}$ maps a simplicial set to its homotopy category hX. The objects are 0-simplices (also called vertexes) of X, and it's arrows are freely generated on 1-simplices with respect to relations governed by 2-simpices.

Chapter 2

Derived functors via deforamtions

I have some familiarity with the beginning material in this section, and will read those parts rather quickly compared to the other parts.

2.1 Homotopical categories and derived functors

The ideas here were introduced around 20 years ago, but we depart slightly from the original terminology. I feel this is important to note to avoid confusions with the literature on the topic.

Definition 2.1 (Homotopical Category)**.** A *homotopical category* is a category $\mathcal{M}$ together with a wide subcategory $\mathcal{W}$ (wide meaning containing all objects) such that for any composable triple f, g, h of arrows, if hg and gf are in $\mathcal{W}$, then $f, g, h, hgf \in \mathcal{W}$. We call arrows in $\mathcal{W}$ *weak equivalences*, and this property is called the 2-of-6 property. ♡

Remark. Weak equivalences in a model category satisfy the 2-of-6 property, so this construction is more general than model categories. However, the construction is not as general as the 2-of-3 property: if two of f, g, gf are in $\mathcal{W}$, then so is the third. This property required for a category to have weak equivalences, and some use this notion to be precisely the same as that of a homotopical category. We distinguish between them and choose to work in categories which satisfy the stronger 2-of-6 property.

Homotopical categories always include all isomorphisms. This is easily proven. Since $\mathcal{W}$ is a subcategory including all objects, it equivalently includes all identities. Consider inverse morphisms $f : a \to b$ and $g : b \to a$, and consider the composable triple (f, g, f). Note $fg = 1_b \in \mathcal{W}$ and $gf = 1_a \in \mathcal{W}$, so we must have $f, g, f, fgf = f, g \in \mathcal{W}$. Conversely, every category admits a minimal collection of weak equivalences whereby it may form a homotopical category, given by the set of isomorphisms in $\mathcal{M}$. In absense of another obvious notion of weak equivalence, we assume the minimal homotopical structure.

Definition 2.2 (Homotopy Category)**.** The homotopy category $\operatorname{Ho}\mathcal{M}$ of a homotopical category $(\mathcal{M}, \mathcal{W})$ is the formal localization of $\mathcal{M}$ at $\mathcal{W}$. ♡

An explicit construction can be given by taking $\operatorname{Obj}(\operatorname{Ho}\mathcal{M}) = \operatorname{Obj}(\mathcal{M})$, and morphisms to be equivalence classes of zig-zags in $\mathcal{M}$[1] with only those in $\mathcal{W}$ permitted to go backwards, quotiented by the relations

- adjacent arrows pointing in the same direction are equal to their composition;

[1] Recall zig-zags are chains of morphisms not necessarily in the composable direction, meaning some are permitted to go backwards

- adjacent pairs $\xleftarrow{w}\xrightarrow{w}$ and $\xrightarrow{w}\xleftarrow{w}$ may be removed;
- identities may be removed.

This defines for each $w \in \mathcal{W}$ a formal inverse w^{-1} which behaves in the expected way. For example, with $\mathcal{W} \subseteq \mathcal{T}\mathrm{op}$ the set of homotopy equivalences, we obtain that for $fg \sim 1$, $g \in [f^{-1}]$.

There is a canonical localization functor $\gamma : \mathcal{M} \to \operatorname{Ho}\mathcal{M}$ mapping $x \mapsto x$ and $f \mapsto [f]$, and it is universal with respect to functors which invert weak equivalences. Hence, precomposition with γ is a bijection between functors $\mathcal{M} \to \mathcal{N}$ which invert weak equivalences and functors $\operatorname{Ho}\mathcal{M} \to \mathcal{N}$.

The homotopy category of the minimal homotopical category is isomorphic to the original category. For another example, a *weak homotopy equivalence* in $\mathcal{T}\mathrm{op}$ is a continuous function $f : X \to Y$ which is an isomorphism of sets of paths, meaning for $U : \mathcal{G}\mathrm{rpoid} \to \mathcal{S}\mathrm{et}$ the forgetful functor $\mathcal{G} \mapsto \mathrm{Obj}(\mathcal{G})$, we have

$$(U\pi_0)(f) : (U\pi_0)(X) \cong (U\pi_0)(Y).$$

Taking weak homotopy equivalences to be weak equivalences, the homotopy category thereof is *equivalent* to the category of CW complexes and homotopy classes of maps.

Remark. In general, not all arrows inverted by localization are weak equivalences. A category for which those arrows inverted are precisely weak equivalences is called *saturated*. Quillen showed all model categories are saturated. We can say slightly more.

Lemma 2.1. *If $\mathcal{M}$ is a category with a collection of arrows $\mathcal{W}$ with a saturated localization, then $(\mathcal{M}, \mathcal{W})$ is homotopical.*

> **Proof.** Saturated means weak equivalences in $\mathcal{M}$ are precisely isomorphisms in $\operatorname{Ho}\mathcal{M}$. Since isomorphisms satisfy the 2-of-6 property, so must the weak equivalences. ■

Annoyingly, homotopy categories need not be locally small. Quillen proved model categories induce locally small localizations, but in the more general setting it is preferable to avoid working directly in the homotopy category to avoid these issues.

Definition 2.3 (Homotopical Functor)**.** A homotopical functor $F : \mathcal{M} \to \mathcal{N}$ between homotopical categories is a functor which preserves weak equivalences. ♡

By univeraslity, homotopical functors induce unique functors

$$\begin{tikzcd}
\mathcal{M} \arrow[r, "F"] \arrow[d, "\gamma"'] & \mathcal{N} \arrow[d, "\delta"] \\
\operatorname{Ho}\mathcal{M} \arrow[r, dashed, "F"'] & \operatorname{Ho}\mathcal{N}
\end{tikzcd}$$

Since we often without loss of generality call both functors F, we say that F commutes with the localizations.

Remark. Let $\mathcal{N}$ be a minimal homotopic category. Universality of localization $\gamma : \mathcal{M} \to \operatorname{Ho}\mathcal{M}$ here is 2-categorical: natural transformations between *homotopical* functors $\mathcal{M} \to \mathcal{N}$ are bijective with natural transformations $\operatorname{Ho}\mathcal{M} \to \mathcal{N}$. If $\mathcal{N}$ is not minimal, we must distinguish between $\operatorname{Ho}\mathcal{N}$ and $\mathcal{N}$. For δ the localization functor, each natural transformation α between functors $\mathcal{M} \to \mathcal{N}$ lowers uniquely to a natural transformation $\delta\alpha$ between functors $\operatorname{Ho}\mathcal{M} \to \operatorname{Ho}\mathcal{N}$ (since $\operatorname{Ho}\mathcal{N}$ inverts weak equivalences and our functors are homotopical). Although this relation is unique, it is not necessarily bijective, and transformations between functors $\operatorname{Ho}\mathcal{M} \to \operatorname{Ho}\mathcal{N}$ can always raise to one between functors $\mathcal{M} \to \operatorname{Ho}\mathcal{N}$, but we may not be able to lift it along δ.

Example. Let $F : \mathcal{A} \to \mathcal{B}$ be a (not necessarily exact) additive functor between abelian categories. Denote by $\mathcal{C}\mathrm{h}_{\geq 0}(\mathcal{A})$ and $\mathcal{C}\mathrm{h}_{\geq 0}(\mathcal{B})$ the categories of chain complexes bounded below at 0. The induced functor $F_\bullet : \mathcal{C}\mathrm{h}_{\geq 0}(\mathcal{A}) \to \mathcal{C}\mathrm{h}_{\geq 0}(\mathcal{B})$ is then homotopical if we take weak equivalences to be *chain complex homotopy equivalences*, but not necessarily if we take them to be quasi-isomorphisms. For example, consider a quasi-isomorphism $f_\bullet : A_\bullet \to B_\bullet$ with $\mathcal{A} = \mathcal{B} = \mathcal{A}\mathrm{b}$, given by

$$\begin{tikzcd}
\dots \arrow[r] & \mathbb{Z}/2\mathbb{Z} \arrow[r, tail] \arrow[d] & \mathbb{Z}/4\mathbb{Z} \arrow[r, "0"] \arrow[d, two heads] & \mathbb{Z}/2\mathbb{Z} \arrow[r, tail] \arrow[d] & \mathbb{Z}/4\mathbb{Z} \arrow[r] \arrow[d, two heads] & 0 \\
\dots \arrow[r] & 0 \arrow[r] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0 \arrow[r] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

This is shown to be a quasi-isomorphism by computing the cohomologies. The computation is not difficult, but takes enough space to type out to make it annoying. We can apply the hom-functor $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, -)$, and we compute

$$\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, 0) = 0, \qquad \operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) = \{0, 1\} = \mathbb{Z}/2\mathbb{Z},$$

$$\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/4\mathbb{Z}) = \{0, (0,1) \mapsto (0,2)\} = \mathbb{Z}/2\mathbb{Z}.$$

Hence we obtain

$$\begin{tikzcd}
\dots \arrow[r] & \mathbb{Z}/2\mathbb{Z}z \arrow[r, "\cong"] \arrow[d] & \mathbb{Z}/2\mathbb{Z} \arrow[r, "0"] \arrow[d, "0"] & \mathbb{Z}/2\mathbb{Z} \arrow[r, "\cong"] \arrow[d] & \mathbb{Z}/2\mathbb{Z} \arrow[r, "0"] \arrow[d, "0"] & 0 \\
\dots \arrow[r] & 0 \arrow[r] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0 \arrow[r] & \mathbb{Z}/2\mathbb{Z} \arrow[r] & 0.
\end{tikzcd}$$

The top row is exact, but note that the homology of the bottom complex at any $\mathbb{Z}/2\mathbb{Z}$ is

$$\frac{\operatorname{Ker}(\mathbb{Z}/2\mathbb{Z} \to 0)}{\operatorname{Im}(0 \to \mathbb{Z}/2\mathbb{Z})} = \frac{\mathbb{Z}/2\mathbb{Z}}{0} = \mathbb{Z}/2\mathbb{Z} \neq 0.$$

Hence, $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, f)_\bullet$ is not a quasi-isomorphism, and additive functors need not be homotopic for this class of weak equivalences.

Many other examples show there are interesting non-homotopical functors. One such class is that of *derived functors*, which are in some sense the closest approximation to homotopical.

Definition 2.4 (Total Derived Functor)**.** A *total left derived functor* $\mathbf{L}F$ of some functor $F : \mathcal{M} \to \mathcal{N}$ for $\mathcal{M}, \mathcal{N}$ homotopic is a *right* Kan extension $\operatorname{Ran}_\gamma \delta F$

$$\begin{tikzcd}
\mathcal{M} \arrow[r, "F"] \arrow[d, "\gamma"'] & \mathcal{N} \arrow[d, "\delta"] \\
\operatorname{Ho}\mathcal{M} \arrow[r, dashed, "\mathbf{L}F"'] \arrow[ur, phantom, "\Uparrow"] & \operatorname{Ho}\mathcal{N}
\end{tikzcd}$$

with γ, δ localization functors. Dually, a *right* derived functor $\mathbf{R}F$ is a *left* Kan extension $\operatorname{Lan}_\gamma \delta F$. ♡

Universality tells us that total derived functors are equivalently homotopical functors $\mathcal{M} \to \operatorname{Ho}\mathcal{N}$, which are often called "derived functors". We reserve that terminology for a special case in which this correspondence is a lift along δ. Sometimes these are called point-set derived functors.

Definition 2.5 (Derived Functor)**.** A *left derived functor* of $F : \mathcal{M} \to \mathcal{N}$ is a homotopical functor $\mathbb{L}F : \mathcal{M} \to \mathcal{N}$ and a natural transformation $\lambda : \mathbb{L}F \Rightarrow F$ such that $\delta \cdot \lambda : \delta \circ \mathbb{L}F \to \delta \circ F$ is a total left derived functor of F (in the sense that it is equal to the Kan extension precomposed with γ). Right derived functors are defined the same. ♡

2.2 Derived functors via deformations

Definition 2.6 (Deformation)**.** A *left deformation* on a homotopical category $\mathcal{M}$ consists of an endofunctor Q and a natural weak equivalence $q : Q \Rightarrow 1_{\mathcal{M}}$. Dually, a right deformation is a natural weak equivalence $q : 1_{\mathcal{M}} \Rightarrow Q$. ♡

Here, that q is a natural weak equivalence means that each component q_m for $m \in \mathcal{M}$ is itself a weak equivalence. Deformations axiomatize a convenient setting in which derived functors exist, and are constructed rather easily. Necessarily Q is homotopical. This is easily shown. Let $w : a \to b$ be a weak equivalence. We then have that $q_b \circ Q(w) = w \circ q_a$. Note that since w and q_a are weak equivalences, their composition must also be a weak equivalence by the 2-of-3 property, which is implied by the 2-of-6 property. Hence, $q_b \circ Q(w)$ is a weak equivalence, and again by the 2-of-3 property we have that $Q(w)$ is a weak equivalence. Now let $\mathcal{M}_Q$ be any full subcategory containing the image of Q. The inclusion $\mathcal{M}_Q \hookrightarrow \mathcal{M}$ and Q define an equivalence of categories between $\operatorname{Ho}\mathcal{M}$ and $\operatorname{Ho}\mathcal{M}_Q$. The book calls $\mathcal{M}_Q$ the subcategory of *cofibrant objects*, and specifies they use the term to mean something different than Quillen's definition in model categories. I will try to abide by another convention and call it a *left deformation retract*.

Definition 2.7 (Deformation for Functors)**.** A *left deformation* for a functor $F : \mathcal{M} \to \mathcal{N}$ between homotopical categories is a left deformation Q for $\mathcal{M}$ such that F is homotopical on a left deforamtion retract $\mathcal{M}_Q \subseteq \mathcal{M}$. We say F is *left deformable* when it admits a left deformation. ♡

In the context of model categories, Ken Brown's lemma shows that a left Quillen functor (a cocontinuous[2] functor preserving cofibrations and trivial cofibrations) will preserve weak equivalences between cofibrant objects (objects where the unique morphism from the intitial object are cofibrations). We prove this lemma later in the text.

As we continue, when we specify a left deformation for a category $\mathcal{M}$, we implicitly choose a left deformation retract.

Theorem 2.1. *If $F : \mathcal{M} \to \mathcal{N}$ has a left deformation $q : Q \Rightarrow 1_{\mathcal{M}}$, then $\mathbb{L}F = F \circ Q$ is a left derived functor of F.*

> **Proof.** Let $\delta : \mathcal{N} \to \operatorname{Ho}\mathcal{N}$ the localization. By definition 2.5, we must show δFQ and $\delta Fq : \delta FQ \Rightarrow \delta F$ form a left Kan extension of δF along γ. Equivalently, we show it satisfies the relevant universal property in $\operatorname{Fun}(\mathcal{M}, \operatorname{Ho}\mathcal{N})$. Let $H : \mathcal{M} \to \operatorname{Ho}\mathcal{N}$ be homotopical (hence meaning it inverts weak equivalences), and $\alpha : G \Rightarrow \delta F$ the relevant natural transformation. Since G is homotopical and q is a natural weak equivalence, $Gq : GQ \Rightarrow G$ is a natural isomorphism. Naturality of α

[2] Conventions appear to vary, this book usees cocontinuous and says some texts require only that (trivial) cofibrations be preserved, but other sources I can find require that it be a left adjoint functor, strengthening the hypothesis.

gives that

$$\delta Fq \circ \alpha Q \circ (Gq)^{-1} = (\delta F \circ \alpha)(q \circ Q) \circ (Gq)^{-1} = \alpha q \circ (Gq)^{-1} = \alpha.$$

Hence, α factors through δFq. To prove uniqueness, first note that since Q is homotopical, qQ is a natural weak equivalence. Since Q is a deformation of F, we have that FqQ is a natural weak equivalence, and thus δFqQ is a natural isomorphism. Let α' be another factorization of α through δFq. We then obtain

$$\begin{tikzcd} GQ \arrow[r, "\alpha' Q"] \arrow[d, "Gq"'] & \delta F Q^2 \arrow[d, "\delta F Q q"] \\ G \arrow[r, "\alpha'"'] & \delta F Q \end{tikzcd}$$

This indeed commutes since Gq is still invertible, so flipping that arrow gives the same equality as above. This indeed also shows the required uniqueness. ■

Hence for $r : 1 \Rightarrow R$ a right deformation and $\mathcal{M}_R$ a right deformation retract, we have also that $\mathbb{R}F = FR$.

Denote by $\mathcal{L}\mathfrak{Def}$ the 2-category of homotopical categories equipped with specified left deformations and left deformation retracts, morphisms given by functors that restrict to homotopical functors on the deformation retracts (left deformable functors with respect to the specified deformation), and 2-morphisms as natural transformations. Note here that a *pseudofunctor* is a functor $\mathcal{Z} \to \mathfrak{Cat}$ which only preserves identities and composition up to coherent natural isomorphism.

Theorem 2.2. *There is a pseudofunctor* $\mathbf{L} : \mathcal{L}\mathfrak{Def} \to \mathfrak{Cat}$ *mapping homotopical categories to their homotopy category, left deformable functors to their total left derived functor, and natural transformations to the derived natural transformations.*

The proof of this theorem (which I omit :3) depends on our construction in theorem 2.1, and it does not hold in generality that the composite of total derived functors is total derived for the composite.

Definition 2.8 (Deformable Adjunction)**.** An adjunction $F \dashv G$ is a *deformable adjunction* if the left adjunct F is left deformable, and the right adjunct G is right deformable. ♡

Theorem 2.3. *If*

$$\begin{tikzcd} \mathcal{M} \arrow[r, shift left, "F"] \arrow[r, phantom, "\perp"] & \mathcal{N} \arrow[l, shift left, "G"] \end{tikzcd}$$

is a deformable adjunction, then so is

$$\mathbf{L}F : \qquad \begin{tikzcd} \operatorname{Ho}\mathcal{M} \arrow[r, shift left] \arrow[r, phantom, "\perp"] & \operatorname{Ho}\mathcal{N} \arrow[l, shift left] \end{tikzcd} \qquad : \mathbf{R}G.$$

Furthermore, the total derived adjunction $\mathbf{L}F \dashv \mathbf{R}G$ *is the unique adjunction compatable with the adjunctions in the sense that*

$$\begin{tikzcd} \mathcal{N}(F(m), n) \arrow[r, phantom, "\cong"] \arrow[d, "\delta"'] & \mathcal{M}(m, G(n)) \arrow[d, "\gamma"] \\ \operatorname{Ho}\mathcal{M}(F(m), n) \arrow[d, "Fq^*"'] & \operatorname{Ho}\mathcal{N}(m, G(n)) \arrow[d, "Gr_*"] \\ \operatorname{Ho}\mathcal{N}(\mathbf{L}F(m), n) \arrow[r, phantom, "\cong"] & \operatorname{Ho}\mathcal{M}(m, \mathbf{R}G(n)) \end{tikzcd}$$

commutes for each $m \in \mathcal{M}$ and $n \in \mathcal{N}$.

The proofs of this fact are very complex, and was omitted from the text. I am not going to go chase after a proof. Instead, I choose to make sense of the statement itself. The only point of confusion is the precomposition by Fq and postcomposition by Gr, but indeed by theorem 2.2 this gives the total derived functor.

Proposition 2.1. *The total left derived functor of a left deformable functor is an absolute Kan extension; in particular, a pointwise right Kan extension.*

Proof. Let $(\delta FQ, \delta Fq)$ be the total left derived functor given by the deformation (Q, q) and theorem 2.2. Let $H : \operatorname{Ho}\mathcal{N} \to \mathcal{E}$. We must show that H preserves the Kan extension $\mathbf{L}F$. To use our construction of $\mathbf{L}F$, we must transfer the universal property to $\mathcal{E}^{\mathcal{M}}$. Let $G : \mathcal{M} \to \mathcal{E}$ homotopical and $\alpha : G \Rightarrow H\delta F$. It suffices to show it factors uniquely though $H\delta Fq$. Similarly to the last proof, α factors as the composition

$$G \xRightarrow{(Gq)^{-1}} GQ \xRightarrow{\alpha Q} H\delta FQ \xRightarrow{H\delta Fq} H\delta F.$$

By the same logic as the above proof, this factorization is unique as well. ■

Exercise. Let $F \dashv G$ and let $\mathbf{L}F$ and $\mathbf{R}G$ be absolute Kan extensions. Show that $\mathbf{L}F \dashv \mathbf{R}G$.

Proof. Let $\eta : 1_{\mathcal{M}} \Rightarrow GF$ be the unit and $\varepsilon : FG \Rightarrow 1_{\mathcal{N}}$ the counit of the adjunction $F \dashv G$. Let $\gamma : \mathcal{M} \to \operatorname{Ho}\mathcal{M}$ and $\delta : \mathcal{N} \to \operatorname{Ho}\mathcal{N}$ the localization functors. Let $\beta : \delta G \Rightarrow \mathbf{R}G$ and $\alpha : \mathbf{L}F \Rightarrow \delta F$ the derived natural transformations. ■

A double category $\mathcal{D}\mathrm{er}$ with objects *deformable categories* (homotopical categories with both left and right deformations, such that either R preserves $\mathcal{M}_Q$ or vice versa), horizontal 1-cells right deformable functors preserving fibrant objects, vertical 1-cells left deformable functors preserving cofibrant objects, and 2-cells appropriate natural transformations. There is then a double pseudofunctor $\mathcal{D} \to \mathcal{C}\mathrm{at}$ given by $\mathcal{C} \mapsto \operatorname{Ho}\mathcal{C}$ and $F \mapsto \mathbf{L}F$; here $\mathcal{C}\mathrm{at}$ is the double category of categories, functors, and natural transformations. No way in hell I'm proving that.

2.3 Classical derived functors between abelian categories

Recall that $\mathcal{M}\mathrm{od}_R$ has enough projectives, meaning for any A there is a bounded below complex of projectives $P_\bullet$ quasi-isomorphic to $\iota(A)$ (a projective resolution of A). Careful choice of projective resolutions is functorial $Q : \mathcal{C}\mathrm{h}_{\geq 0}(R) \to \mathcal{C}\mathrm{h}_{\geq 0}(R)$ with a natural quasi-isomorphism $q : Q \Rightarrow 1$. Dually, modules admit injective resolutions which give a right deformation $r : 1 \Rightarrow (R : \mathcal{C}\mathrm{h}^{\geq 0}(R) \to \mathcal{C}\mathrm{h}^{\geq 0}(R))$ on the category of bounded below cochain complexes.

Additive functors $F : \mathcal{M}\mathrm{od}_R \to \mathcal{M}\mathrm{od}_S$ induce functors $F_\bullet : \mathcal{C}\mathrm{h}_{\geq 0}(R) \to \mathcal{C}\mathrm{h}_{\geq 0}(S)$ andd $F^\bullet : \mathcal{C}\mathrm{h}^{\leq 0}(R) \to \mathcal{C}\mathrm{h}^{\leq 0}(S)$ preserving chain homotopies. When working with projective/injective resolutions, the quasi-isomorphisms become homotopy equivalences, and are thus preserved by $F_\bullet$ and $F^\bullet$. Hence, with respect to quasi-isomorphisms, $F^\bullet$ has a right derived functor and $F_\bullet$ has a left derived functor, as described herein. The functor that we classically refer to as the left derived functor of $F : \mathcal{M}\mathrm{od}_R \to \mathcal{M}\mathrm{od}_S$ in homological algebra is the composition

$$\mathcal{M}\mathrm{od}_R \xrightarrow{\iota} \mathcal{C}\mathrm{h}_{\geq 0}(R) \xrightarrow{\mathbb{L}F_\bullet} \mathcal{C}\mathrm{h}_{\geq 0}(S) \xrightarrow{H_0} \mathcal{M}\mathrm{od}_S.$$

One can also take the other derived functors, but the derivations we chose induce long exact sequences, so we prefer those.

2.4 Preview of homotopy limits and colimits

Let $\mathcal{M}$ a complete and cocomplete homotopical category, and $\mathcal{D}$ a category indexing some small diagram. The colimit and limit functors $\varprojlim, \varinjlim : \mathcal{M}^{\mathcal{D}} \to \mathcal{M}$ need not be homotopical with respect to pointwise weak equivalences in $\mathcal{M}^{\mathcal{D}}$ (I assume meaning natural weak equivalences). However, since the functors are continuous, their respective Kan extensions exist, so we would want to define a left derived functor of $\varinjlim$ and a right derived functor of $\varprojlim$, called hocolim and holim, respectively. Unfortunately, even if $\mathcal{M}$ is a model category, $\mathcal{M}^{\mathcal{D}}$ may not have a model structure such that $\varinjlim$ and $\varprojlim$ are Quillen. Hence the theory of deformations exists for construction derived functors without invoking model structures.

Chapter 3

Basic concepts of enriched category theory

There are many reasons to consider enrichments, but one notable one for this text is the desire for a "homotopy product", which would be a product diagram that only commutes *up to homotopy*. In order to have some notion of homotopy for which this makes sense, we can assume hom-sets are equipped with topologies, and then homotopies between arrows would become paths in the underlying space. A homotopy product can then be equipped with a natural weak homotopy equivalence.

Since we have not defined a class of weak equivalences, the homotopy category $h\mathcal{M}$ is different from the ones we worked with earlier. Denoting by $\underline{\mathcal{M}}(a, b)$ the space assigned to $\mathcal{M}(a, b)$, we define $h\mathcal{M}(a, b) \coloneqq \pi_0\underline{\mathcal{M}}(a, b)$. Here note that π_0 is the functor $\mathcal{T}\mathrm{op} \to \mathcal{S}\mathrm{et}$ assigning to each space it's path components; hence taking a space X and quotienting in $\mathcal{S}\mathrm{et}$ via the relation $x \sim y$ iff there is a path from x to y.

3.1 A first example

Let $\mathcal{M}\mathrm{od}_R$ be the category of left R-modules. We well know that the set $\mathcal{M}\mathrm{od}_R(A, B)$ admits an abelian group structure via pointwise addition, which we will denote by $\underline{\mathcal{M}\mathrm{od}_R}(A, B)$. Moreover, composition

$$\circ : \mathcal{M}\mathrm{od}_R(B, C) \times \mathcal{M}\mathrm{od}_R(A, B) \to \mathcal{M}\mathrm{od}_R(A, C)$$

distributes over addition, and hence is $\mathbb{Z}$-bilinear. Thus it factors through $\otimes_{\mathbb{Z}}$, inducing an abelian group homomorphism

$$\underline{\mathcal{M}\mathrm{od}_R}(B, C) \otimes_{\mathbb{Z}} \underline{\mathcal{M}\mathrm{od}_R}(A, B) \to \underline{\mathcal{M}\mathrm{od}_R}(A, C)$$

in the category $\mathcal{A}\mathrm{b}$.

Note also that since any abelian group a has $\mathcal{A}\mathrm{b}(\mathbb{Z}, a) \cong a$, we can represent the identity morphism of A as an abelian group homomorphism

$$1_A : \mathbb{Z} \to \underline{\mathcal{M}\mathrm{od}_R}(A, A).$$

Moreover, $\mathbb{Z}$ is a unit for the monoidal product $\otimes_{\mathbb{Z}} : \mathcal{A}\mathrm{b} \times \mathcal{A}\mathrm{b} \to \mathcal{A}\mathrm{b}$ via left and right natural isomorphisms. This allopws us to encode the usual composition axioms via diagrams in $\mathcal{A}\mathrm{b}$. These facts show that $\mathcal{M}\mathrm{od}_R$ admits the structure of a category enriched over $(\mathcal{A}\mathrm{b}, \otimes_{\mathbb{Z}}, \mathbb{Z})$.

3.2 The base for enrichment

A symmetric monoidal category $(\mathcal{V}, \times, *)$ is a category with a bifunctor $- \times - : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$ called the monoidal product, and an element $* \in \mathcal{V}$ called the unit, together with specified natural isomorphisms

$$v \times w \cong w \times v, \qquad u \times (v \times w) \cong (u \times v) \times w, \qquad * \times v \cong v \cong v \times *$$

satisfying coherence criteria. Any category with finite products is symmetric monoidal with terminal object the unit; we call a category with such a monoidal structure *cartesian monoidal*. In most relevant examples, $\mathcal{V}$ is complete and cocomplete, so we will generally assume this.

3.3 Enriched categories

Definition 3.1 (Enriched Category)**.** Let $\mathcal{V}$ be a symmetric[1] monoidal category. A $\mathcal{V}$-category $\underline{\mathcal{D}}$ is

- A collection of objects $x, y, z \in \underline{\mathcal{D}}$;
- For each pair $x, y \in \underline{\mathcal{D}}$, a hom-*object* $\underline{\mathcal{D}}(x, y) \in \mathcal{V}$;
- For each $x \in \underline{\mathcal{D}}$, a morphism $\mathrm{id}_x : * \to \underline{\mathcal{D}}(x, x)$ in $\mathcal{V}$;
- For each triple $x, y, z \in \underline{\mathcal{D}}$, a morphism $\circ : \underline{\mathcal{D}}(y, z) \otimes \underline{\mathcal{D}}(xy) \to \underline{\mathcal{D}}(x, z)$ in $\mathcal{V}$;

such that for all $w, x, y, z \in \underline{\mathcal{D}}$, the diagrams

$$\begin{tikzcd}
\underline{\mathcal{D}}(z,y) \otimes \underline{\mathcal{D}}(x,y) \otimes \underline{\mathcal{D}}(w,x) \arrow[r, "1\otimes\circ"] \arrow[d, "\circ\otimes 1"'] & \underline{\mathcal{D}}(z,y) \otimes \underline{\mathcal{D}}(x,z) \arrow[d, "\circ"] \\
\underline{\mathcal{D}}(y,w) \otimes \underline{\mathcal{D}}(w,x) \arrow[r, "\circ"'] & \underline{\mathcal{D}}(w,z)
\end{tikzcd}$$

$$\begin{tikzcd}
\underline{\mathcal{D}}(x,y) \otimes * \arrow[r, "1\otimes \mathrm{id}_x"] \arrow[dr, "\cong"'] & \underline{\mathcal{D}}(x,y) \otimes \underline{\mathcal{D}}(x,x) \arrow[d, "\circ"] \\
& \underline{\mathcal{D}}(x,y)
\end{tikzcd}
\qquad
\begin{tikzcd}
\underline{\mathcal{D}}(y,y) \otimes \underline{\mathcal{D}}(x,y) \arrow[d, "\circ"] & * \otimes \underline{\mathcal{D}}(x,y) \arrow[l, "\mathrm{id}_y \otimes 1"'] \arrow[dl, "\cong"] \\
\underline{\mathcal{D}}(x,y) &
\end{tikzcd}$$

commute in $\mathcal{V}$. ♡

We omit a relevant associativity morphism in the first diagram because the text did it too and I only realized after copying it all down and I'm NOT editing it.

We already know of a plethora of examples. For example, a ring is an $\mathcal{A}b$-category with a singleton for objects. A more interesting example is that topological spaces are *naturally* enriched over groupoids. Define $\underline{\mathcal{T}\mathrm{op}}(X, Y)$ to be the groupoid with objects as continuous maps $X \to Y$ and morphisms as homotopy classes of homotopies between these maps.

If $\mathcal{V}$ has copowers of the unit that are preserved by tensor in the following sense, then any unenriched $\mathcal{C}$ has a free $\mathcal{V}$-category with objects the same as $\mathcal{C}$. Hom-objects from a to b are given by the copower $\coprod_{\mathcal{C}(a,b)} *$. Identity morphisms id_a are given by the inclusion of $*$ into the copower $\coprod_{\mathcal{C}(a,a)} *$ at the identity 1_a. Since tensor preserves the relevant copower, we obtain

$$\left(\coprod_{\mathcal{C}(b,c)} *\right) \otimes \left(\coprod_{\mathcal{C}(a,b)} *\right) \cong \coprod_{\mathcal{C}(b,c)} \left(* \otimes \left(\coprod_{\mathcal{C}(a,b)} *\right)\right) \cong \coprod_{\mathcal{C}(b,c)} \left(\coprod_{\mathcal{C}(a,b)} (* \otimes *)\right) \cong \coprod_{\mathcal{C}(b,c)} \coprod_{\mathcal{C}(a,b)} * \cong \coprod_{\mathcal{C}(b,c) \times \mathcal{C}(a,b)} *.$$

Since there is a composition function $\mathcal{C}(b, c) \times \mathcal{C}(a, b)$, composition in the free $\mathcal{V}$-category simply reindexes along this arrow.

---

[1] We are using symmetric monoidal categories here, but the symmetric hypothesis can be dropped for general enrichments. We require it for some results, however. I will assume we only enrich over symmetric monoidal categories unless the text specifies otherwise.

Definition 3.2 (Closed Monoidal Category)**.** Let $\mathcal{V}$ be a symmetric monoidal category. If each functor $- \otimes v : \mathcal{V} \to \mathcal{V}$ has a right adjoint, denoted $\underline{\mathcal{V}}(v, -)$, the right adjoints assemble uniquely into a bifunctor

$$\underline{\mathcal{V}}(-,-) : \mathcal{V}^{\mathrm{op}} \times \mathcal{V} \to \mathcal{V}$$

such that for all $u, v, w \in \mathcal{V}$,

$$\mathcal{V}(u \times v, w) \cong \mathcal{V}(u, \underline{\mathcal{V}}(v, w))$$

natural in u, v, w.

We call $\underline{\mathcal{V}}(-,-)$ the *internal hom*, and $\mathcal{V}$ is enriched over itself with composition

$$\underline{\mathcal{V}}(v, w) \otimes \underline{\mathcal{V}}(u, v) \to \underline{\mathcal{V}}(u, w)$$

given as follows. Note the adjunction $- \otimes u \dashv \underline{\mathcal{V}}(u, -)$ gives the counit

$$\varepsilon^u : (- \otimes u) \circ \underline{\mathcal{V}}(u, -) = \underline{\mathcal{V}}(u, -) \otimes u \Rightarrow 1.$$

Thus the adjunctions give a composition

$$\underline{\mathcal{V}}(v, w) \otimes \underline{\mathcal{V}}(u, v) \otimes u \xrightarrow{1 \otimes \varepsilon^u_v} \underline{\mathcal{V}}(v, w) \otimes v \xrightarrow{\varepsilon^v_w} w.$$

Hence we have

$$\varepsilon^v_w \circ (\varepsilon^u_v \otimes 1) \in \mathcal{V}\left(\underline{\mathcal{V}}(v, w) \otimes \underline{\mathcal{V}}(u, v) \otimes u, w\right) \cong \mathcal{V}\left(\underline{\mathcal{V}}(v, w) \otimes \underline{\mathcal{V}}(u, v), \underline{\mathcal{V}}(u, w)\right).$$

The isomorphism follows by applying symmetry to the adjunction. We thus define composition as the adjunct of the relevant composition. The identity morphisms are defined as follows. The natural isomorphism $\lambda_v : * \otimes v \cong v$ is adjunct under the same adjunction to the identity id_v:

$$\lambda \in \mathcal{V}(* \otimes v, v) \cong \mathcal{V}(*, \underline{\mathcal{V}}(v, v)).$$

Such a triple $(\mathcal{V}, \otimes, *)$ is a *closed symmetric monoidal category*. ♡

When $\mathcal{V}$ is cartesian monoidal, we call it *cartesian closed*. An example is $\mathrm{s}\mathcal{S}\mathrm{et}$. The product is the usual product of functors, and we desire an adjunction

$$\mathrm{s}\mathcal{S}\mathrm{et}(X \times Y, Z) \cong \mathrm{s}\mathcal{S}\mathrm{et}(X, Z^Y).$$

Here, we use exponential notation $\underline{\mathrm{s}\mathcal{S}\mathrm{et}}(X, Y) =: Y^X$. If X is representable $\mathbf{\Delta}(-, [n])$, which we write as Δ^n, then Yoneda gives

$$(Z^Y)_n \cong \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, Z^Y) \cong \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n \times Y, Z).$$

Hence we obtain a valid definition of Z^Y by setting the n-simplices as the set of maps $\Delta^n \times Y \to Z$. Hence $- \times Y \dashv (-)^Y$ for each simplicial set Y, giving that $\mathrm{s}\mathcal{S}\mathrm{et}$ is a closed cartesian monoidal category.

The category $\mathcal{C}\mathrm{at}$ is cartesian closed, since functors and natural transformations between 2 fixed categories form another category. The precise definition of a 2-category is a category enriched over $\mathcal{C}\mathrm{at}$. Objects are 0-cells, the 1-cells are the objects of the hom-categories, and the 2-cells are the morphisms in the hom-categories. This interpretation gives us a notion of arrows in the enriched category, compatable with the notion of composition given in definition 3.2. Equivalently, we can take 1-cells to be the arrows of the *underlying category*, defined later.

The category $\mathcal{T}\mathrm{op}$ is *not* cartesian closed, but several large convenient subcategories are, such as $\mathcal{CGH}\mathrm{aus}$. For now, $\mathcal{T}\mathrm{op}$ will denote one of these convenient categories, and we will make precise which categories we concern ourselves with later in the text.

We provide the following construction. Let $(\mathcal{V}, \times, *)$ be a cartesian closed symmetric monoidal category, which we also assume to be complete and cocomplete. There is a general procedure for producing a closed symmetric monoidal category $(\mathcal{V}_*, \wedge, S^0)$ of *pointed objects* or *based objects*, equivalently the slice category $(* \downarrow \mathcal{V}) =: \mathcal{V}_*$. We define $S^0 := * \amalg *$ (the coproduct in $\mathcal{V}$). Note that for each v, w, we can define a map $v \to w$ by factoring through the terminal object $v \to * \to w$ via the basepoint on w. This gives maps $v, w \to v \times w$, and universality of coproduct gives a canonical map $f : v \amalg w \to v \times w$.We then define the *smash product* $\wedge$ to be the pushout in $\mathcal{V}$:

$$\begin{tikzcd}
v \amalg w \arrow[r, "f"] \arrow[d] & v \times w \arrow[d] \arrow[ddr, bend left] & \\
* \arrow[r] \arrow[drr, bend right] & v \wedge w \arrow[dr, dashed] & \\
& & z.
\end{tikzcd}$$

We then define the hom-objects $\underline{\mathcal{V}}_*$ in $\mathcal{V}_*$ to be pullbacks of internal homs in $\mathcal{V}$:

$$\begin{tikzcd}
z \arrow[dr, dashed] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
& \underline{\mathcal{V}}_*(v, w) \arrow[r] \arrow[d] & \underline{\mathcal{V}}(v, w) \arrow[d] \\
& \underline{\mathcal{V}}(*, *) \arrow[r] & \underline{\mathcal{V}}(*, w).
\end{tikzcd}$$

The bottom map is postcomposition by the base map $* \to w$, and the downwards map is precomposition by the basepoint $* \to v$. This does not have a precise definition yet, but intuitively it makes sense, since hom-objects in cartesian monoidal categories are intuitively similar to hom-sets. There is then a forgetful functor $U : \mathcal{V}_* \to \mathcal{V}$ forgetting the basepoint, which is left adjoint to the functor $(-)_+ : \mathcal{V} \to \mathcal{V}_+$ given by computing the coproduct with the terminal object.

Lemma 3.1. *When $\mathcal{V}$ is a cartesian closed symmetric monoidal category, the functor $(-)_+ : \mathcal{V} \to \mathcal{V}_*$ is* strong monoidal, *meaning*

$$S^0 \cong (*)_+, \qquad v_+ \wedge w_+ \cong (v \times w)_+$$

with these isomorphisms natural, and appropriately commuting with associators and unitors.

Proof. The unit isomorphism is by definition of S^0. By definition, the object $v_+ \wedge w_+$ is given by the cofiber (the opposite arrow in the pushforward diagram) of

$$v_+ \amalg w_+ \to v_+ \times w_+ = (v \amalg *) \times (w \amalg *).$$

Since $\mathcal{V}$ is cartesian closed and symmetric, $v \times -$ and $- \times v$ are left-adjoint and thus preserve colimits, and thus preserve coproducts. Thus, $\times$ distributes over $\amalg$, and we obtain

$$(v \amalg *) \times (w \amalg *) \cong (v \times (w \amalg *)) \amalg (* \times (w \amalg *)) \cong ((v \times w) \amalg (v \times *)) \amalg ((* \times w) \amalg (* \times *)).$$

Dropping the unnecessary parenthases due to associativity of $\amalg$, we also have

$$(v \times w) \amalg (v \times *) \amalg (* \times w) \amalg (* \times *) \cong (v \times w) \amalg v \amalg w \amalg *$$

since the terminal object $*$ is a unit for $\times$. Universality of coproduct gives an arrow

$$v_+ \amalg w_+ \to v \amalg w \amalg *$$

which the canonical arrow $v_+ \amalg w_+ \to v_+ \times w_+$ factors through by uniqueness. Hence we obtain a diagram

$$\begin{tikzcd} v_+ \amalg w_+ \arrow[r, two heads] \arrow[d] & v \amalg w \amalg * \arrow[r] & v \times w \amalg v \amalg w \amalg * \arrow[d] \\ * \arrow[r] & * \arrow[r] & v \times w \amalg * = (v \times w)_+ \end{tikzcd}$$

since the canonical map doesn't act on $v \times w$, which is a double pushout, since the top left arrow is epi. ■

A convenient category $\mathcal{T}\mathrm{op}$ of spaces gives a closed symmetric monoidal category $(\mathcal{T}\mathrm{op}_*, \wedge, S^0)$. So does $(\mathrm{s}\mathcal{S}\mathrm{et}_*, \wedge, \partial\Delta^1)$.

3.4 Underlying categories of enriched categories

Here is some motivation which I'm copying down less for the motivation, and more because I think the new concepts are cool. Fix a group G to which the discrete topology has been associated, and take $\mathcal{T}\mathrm{op}$ some well-behaved category of topological spaces. The functor category $\mathcal{T}\mathrm{op}^G$ has as objects G-spaces and as morphisms G-equivariant maps. Unpacking this quickly, a functor is a group action of G on a topological space, meaning a choice of continuous automorphisms of a space that respect the group operation. Morphisms of G-spaces are natural transformations, but since grououps are singletons, this amounts to a single continuous map that preserves the group action, called G-equivariant.

This category admits two natural enrichments. First, the set of G-equivariant maps can be viewed as a subspace $\underline{\mathcal{T}\mathrm{op}}^G(X,Y) \subseteq \underline{\mathcal{T}\mathrm{op}}(X,Y)$, where the latter has the *compact-open topology* generated by sets

$$V(K,U) \coloneqq \{f \in \mathcal{T}\mathrm{op}(X,Y) \mid f(K) \subseteq U\}, \qquad K \subseteq X \text{ compact}, \qquad U \subseteq Y \text{ open}.$$

This gives a topological enrichment.

Additionally, $\mathcal{T}\mathrm{op}^G$ is symmetric cartesian monoidal, with product given by the standard cartesian product together with the diagonal action $(g,h)(x,y) = (gx, hy)$. The singleton with the trivial action is a unit thereof. The space $\underline{\mathcal{T}\mathrm{op}}(X,Y)$ has a canonical G-action

$$g \cdot f(x) = gf(g^{-1}x).$$

This gives an enrichment over $\mathcal{T}\mathrm{op}^G$, and we write $\underline{\mathcal{T}\mathrm{op}}_G(X,Y)$ for this space to distinguish from the other enrichment. With this definition $\mathcal{T}\mathrm{op}^G$ is cartesian closed.

We think of $\mathcal{T}\mathrm{op}^G$ as the underlying (unenriched) category for both categories, but this intuition is difficult to formalize in full generality. However, generality in this case helps extract the full definition.

To define an underlying category, we first need a way to construct underlying sets, and hence a functor $\mathcal{V} \to \mathcal{S}\mathrm{et}$. For example, in $\mathcal{A}\mathrm{b}$, $\mathrm{s}\mathcal{S}\mathrm{et}$, $\mathcal{T}\mathrm{op}$, $\mathcal{C}\mathrm{at}$, and more, the underlying set functor is simply the representable functor $\mathcal{V}(*, -) : \mathcal{V} \to \mathcal{S}\mathrm{et}$, since $*$ is a singleton. Since this is the unit for the cartesian monoidal structures of these categories, we might generalize this to any symmetric monoidal category with $*$ as the unit. This is on the right track, but we need a bit more.

Definition 3.3 (Lax Monoidal)**.** A functor $F : \mathcal{V} \to \mathcal{U}$ between symmetric monoidal categories $(\mathcal{V}, \otimes, *)$ and $(\mathcal{U}, \square, 1)$ is *lax monoidal* if there are natural transformations (NOT ISOMORPHISMS)

$$F(v) \otimes F(v') \to F(v \,\square\, v'), \qquad 1 \to F(*).$$

♡

Clearly all strong monoidal functors are lax, with natural transformations as isomorphisms.

Lemma 3.2 (Change of Base)**.** *Let $F : \mathcal{V} \to \mathcal{U}$ be lax monoidal. Then any $\mathcal{V}$-category has an associated $\mathcal{U}$-category with the same objects.*

Proof. Let $\underline{\mathcal{D}}_{\mathcal{V}}$ be a $\mathcal{V}$-category. Define a $\mathcal{U}$-category $\underline{\mathcal{D}}_{\mathcal{U}}$ with the same objects, and with hom-objects $\underline{\mathcal{D}}_{\mathcal{U}}(x, y) \coloneqq F\underline{\mathcal{D}}_{\mathcal{V}}(x, y)$. The lax monoidal structure of F gives composition and unit maps

$$F\underline{\mathcal{D}}_{\mathcal{V}}(y, z) \,\square\, F\underline{\mathcal{D}}_{\mathcal{V}}(x, y) \to F\left(\underline{\mathcal{D}}_{\mathcal{V}}(y, z) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(x, y)\right) \xrightarrow{F(\circ)} F\underline{\mathcal{D}}_{\mathcal{V}}(x, z) = \underline{\mathcal{D}}_{\mathcal{U}}(x, z);$$

$$1 \to F(*) \xrightarrow{F(\mathrm{id}_x)} F\underline{\mathcal{D}}_{\mathcal{V}}(x, x) = \underline{\mathcal{D}}_{\mathcal{U}}(x, x).$$

■

The point is that $\mathcal{V}(*, -)$ is lax monoidal. Let $v, w \in \mathcal{V}$. Then the bifunctor $\otimes$ gives for any pair $* \to v$ and $* \to w$ a map $* \otimes * \to v \otimes w$. Precomposing with the $\otimes$-diagonal map, indeed an isomorphism here, gives rise to a function

$$\mathcal{V}(*, v) \times \mathcal{V}(*, w) \to \mathcal{V}(*, v \otimes w)$$

natural in v and w, and hence $\mathcal{V}(*, -)$ is lax functorial. We then apply lemma 3.6 to change the base for the enrichment from $\mathcal{V}$ to $\mathcal{S}\mathrm{et}$.

Definition 3.4 (Underlying Category)**.** Given a $\mathcal{V}$-category $\underline{\mathcal{C}}$, the underlying category $\mathcal{C}_0$ has the same objects, and as hom-sets

$$\mathcal{C}_0(x, y) \coloneqq \mathcal{V}(*, \underline{\mathcal{C}}(x, y)).$$

Definitions are as in lemma 3.6: $1_x = \mathrm{id}_x \in \mathcal{V}(*, \underline{\mathcal{C}}(x, x))$ and

$$\begin{tikzcd}
\mathcal{C}_0(y, z) \times \mathcal{C}(x, y) \arrow[rr, dashed, "\circ"] \arrow[d, equal] & & \mathcal{C}_0(x, z) \arrow[d, equal] \\
\mathcal{V}(*, \underline{\mathcal{C}}(y, z)) \times \mathcal{V}(*, \underline{\mathcal{C}}(x, y)) \arrow[r, "\otimes"'] & \mathcal{V}(*, \underline{\mathcal{C}}(y, z) \otimes \underline{\mathcal{C}}(x, y)) \arrow[r, "\mathcal{V}(*, \circ)"'] & \mathcal{V}(*, \underline{\mathcal{C}}(x, z)).
\end{tikzcd}$$

♡

Reconsiling our previous example, first consider the $\mathcal{T}\mathrm{op}$-enriched category $\underline{\mathcal{T}\mathrm{op}}^G$. The hom-sets of the underling category are $\mathcal{T}\mathrm{op}(*, \underline{\mathcal{T}\mathrm{op}}^G(X,Y))$, equivalently the set of points in $\underline{\mathcal{T}\mathrm{op}}^G(X,Y)$, equivalently all G-equivariant maps $X \to Y$. Hence the underlying category is $\mathcal{T}\mathrm{op}^G$. Similarly, the hom-sets of the underlying category of the $\mathcal{T}\mathrm{op}^G$-enriched category $\underline{\mathcal{T}\mathrm{op}}_G$ are given by $\mathcal{T}\mathrm{op}^G(*, \underline{\mathcal{T}\mathrm{op}}_G(X,Y))$. Since $*$ is the trivial group action, G-equivariant maps $* \to \underline{\mathcal{T}\mathrm{op}}_G(X,Y)$ are G-fixed points. Note that fixed points for the conjugation action are precisely G-equivariant maps:

$$g \cdot f(x) = gf(g^{-1}x) = gg^{-1}f(x) = f(x).$$

Hence, the underlying category is once again $\mathcal{T}\mathrm{op}^G$.

These considerations with group actions generalize. For $\mathcal{V}$ a complete and cocomplete closed symmetric monoidal category, $\mathcal{V}^G$ inherits a canonical closed symmetrical monoidal structure such that the forgetful functor $U : \mathcal{V}^G \to \mathcal{V}$ preserves the unit, tensor, and internal homs. The hom-object $\underline{\mathcal{V}}^G(c,d)$ agrees with $\underline{\mathcal{V}}(c,d)$, and the underlying category is again the original $\mathcal{V}^G$. The discussion also generalizes in another way.

Lemma 3.3. *If $(V, \otimes, *)$ is a closed symmetric monoidal category, the underlying category of the $\mathcal{V}$-category $\underline{\mathcal{V}}$ is $\mathcal{V}$.*

Proof. Internal homs give the adjunction

$$\mathcal{V}(*, \underline{\mathcal{V}}(x,y)) \cong \mathcal{V}(* \otimes x, y) \cong \mathcal{V}(x,y),$$

with the second isomorphism the unitor. Hence, hom-sets of the underlying category agree with $\mathcal{V}$. We now need to show identities and composition agree. The first one is handled by functorality of $\mathcal{V}(*, -)$, so now we need only show composition agrees.

Using this above isomorphism, we take $f : x \to y$ in $\mathcal{V}$ and the morphism $f : * \to \underline{\mathcal{V}}(x,y)$ to be equivalent, hence we use the same label. Let $\varepsilon_y^x : \underline{\mathcal{V}}(x,y) \otimes x \to y$ be the relevant counit of the relevant adjunction. Following the isomorphism gives a precise definition of $f : x \to y$ in terms of $f : * \to \underline{\mathcal{V}}(x,y)$ as the composition

$$x \cong * \otimes x \xrightarrow{f \otimes 1} \underline{\mathcal{V}}(x,y) \otimes x \xrightarrow{\varepsilon_y^x} y.$$

Now we follow composition as in definition 3.4. The composition in $\underline{\mathcal{V}}$ is defined as

$$* \cong * \otimes * \xrightarrow{g \otimes f} \underline{\mathcal{V}}(y,z) \otimes \underline{\mathcal{V}}(x,y) \xrightarrow{\circ} \underline{\mathcal{V}}(x,z).$$

Recall composition was adjunct to

$$\underline{\mathcal{V}}(y,z) \otimes \underline{\mathcal{V}}(x,y) \otimes x \xrightarrow{1 \otimes \varepsilon_y^x} \underline{\mathcal{V}}(y,z) \otimes y \xrightarrow{\varepsilon_z^y} z.$$

Naturality and precomposition with a unitor gives that our definition of composition in its entirety is adjunct to

$$* \otimes x \cong * \otimes * \otimes x \xrightarrow{g \otimes f \otimes 1} \underline{\mathcal{V}}(y,z) \otimes \underline{\mathcal{V}}(x,y) \otimes x \xrightarrow{1 \otimes \varepsilon_y^x} \underline{\mathcal{V}}(y,z) \otimes y \xrightarrow{\varepsilon_z^y} z.$$

Functorality and another unitor gives

$$x \cong * \otimes * \otimes x \xrightarrow{g\otimes 1\otimes 1} \underline{\mathcal{V}}(y,z) \otimes * \otimes x \xrightarrow{1\otimes f\otimes 1} \underline{\mathcal{V}}(y,z) \otimes \underline{\mathcal{V}}(x,y) \otimes x \xrightarrow{1\otimes \varepsilon^x_y} \underline{\mathcal{V}}(y,z) \otimes y \xrightarrow{\varepsilon^y_z} z.$$

By functorality of tensor, we can replace the part of the composition

$$* \otimes x \xrightarrow{f\otimes 1} \underline{\mathcal{V}}(x,y) \otimes x \xrightarrow{\varepsilon^x_y} y$$

with the definition of $f : x \to y$ in $\mathcal{V}$:

$$x \cong * \otimes x \xrightarrow{g\otimes 1} \underline{\mathcal{V}}(y,z) \otimes x \xrightarrow{1\otimes f} \underline{\mathcal{V}}(y,z) \otimes y \xrightarrow{\varepsilon^y_z} z.$$

Functorality of $\otimes$ gives that this composition is equivalent to

$$x \xrightarrow{f} y \cong * \otimes y \xrightarrow{g\otimes 1} \underline{\mathcal{V}}(y,z) \otimes y \xrightarrow{\varepsilon^y_z} z.$$

The tail of this composition is precisely the definition of g as an arrow in the underlying category, hence this is the composition $g \circ f$. ■

Exercise. Let $\underline{\mathcal{D}}$ be a $\mathcal{V}$-category and $g : * \to \underline{\mathcal{D}}(y,z)$ an arrow in $\mathcal{D}_0$. Define for any $x \in \underline{\mathcal{D}}$ postcomposition:

$$g_* : \underline{\mathcal{D}}(x,y) \cong * \otimes \underline{\mathcal{D}}(x,y) \xrightarrow{g\otimes 1} \underline{\mathcal{D}}(y,z) \otimes \underline{\mathcal{D}}(x,y) \xrightarrow{\circ} \underline{\mathcal{D}}(x,z).$$

Precomposition is defined similarly. This definition gives an (unenriched) functor

$$\underline{\mathcal{D}}(x,-) : \mathcal{D}_0 \to \mathcal{V}, \qquad y \mapsto \underline{\mathcal{D}}(x,y), \qquad g \mapsto g_*.$$

Show that the composition with the underlying set functor $\mathcal{V}(*,-) \circ \underline{\mathcal{D}}(x,-)$ is the representable functor $\mathcal{D}_0(x,-) : \mathcal{D}_0 \to \mathcal{S}\mathrm{et}$.

Proof. By definition 3.4, $\mathcal{V}(*, \underline{\mathcal{D}}(x,-)) =: \mathcal{D}_0(x,-)$. Hence, the definitions agree on objects. We also have that $\mathcal{D}_0(x,g) = g_*$ in the conventional sense of the notation, and by definition $\underline{\mathcal{D}}(x,g) = g_*$ in the sense of the notation as above. We also have that for a morphism f in $\mathcal{V}$, $\mathcal{V}(x,f)$ is also precomposition in the conventional sense. Note that $\mathcal{V}(*, \underline{\mathcal{D}}(x,g)) = \mathcal{V}(*, g_*)$.

Let $f : x \to y$ in $\mathcal{D}_0$, and equivalently $f : * \to \underline{\mathcal{D}}(x,y)$ in $\mathcal{V}$. We must show that evaluating $\mathcal{V}(*, g_*)(f)$ is precisely the same as $\mathcal{D}_0(x,g)(f)$. Since g_* is defined as an arrow in $\mathcal{V}$, the former is given as $g_* \circ f$ with composition as defined for $\mathcal{V}$. The latter is given as $g \circ f$, with composition as defined in $\mathcal{D}_0$. Writing out the composition $g_* \circ f$ as defined above gives the composition

$$* \xrightarrow{f} \underline{\mathcal{D}}(x,y) \cong * \otimes \underline{\mathcal{D}}(x,y) \xrightarrow{g\otimes 1} \underline{\mathcal{D}}(y,z) \otimes \underline{\mathcal{D}}(x,y) \xrightarrow{\circ} \underline{\mathcal{D}}(x,z)$$

with composition as in $\underline{\mathcal{D}}$. Functorality simplifies this to

$$* \cong * \otimes * \xrightarrow{g\otimes f} \underline{\mathcal{D}}(y,z) \otimes \underline{\mathcal{D}}(x,y) \xrightarrow{\circ} \underline{\mathcal{D}}(x,z).$$

By definition 3.4, this is precisely the same as the definition of $g \circ f$ in $\mathcal{D}_0$. Thus, $\mathcal{V}(*, \underline{\mathcal{D}}(x, -)) = \mathcal{D}_0(x, -)$. ∎

Hence we have a good notion of precomposition and postcomposition. Importantly, pre- and postcomposition are continuous in topological categories, but we can indeed show more. These representable functors are $\mathcal{V}$-functors.

3.5 Enriched functors and enriched natural transformations

Small $\mathcal{V}$-categories can form a 2-category (if $\mathcal{V}$ is allowed to be symmetric). To make precise this notion we require the notion of a $\mathcal{V}$-functor.

Definition 3.5 ($\mathcal{V}$-Functor)**.** A $\mathcal{V}$-functor $F : \underline{\mathcal{C}} \to \underline{\mathcal{D}}$ between $\mathcal{V}$-categories is an object map $x \mapsto F(x)$ from $\underline{\mathcal{C}}$ to $\underline{\mathcal{D}}$, together with morphisms

$$F_{x,y} : \underline{\mathcal{C}}(x, y) \to \underline{\mathcal{D}}(x, y)$$

in $\mathcal{V}$ for each pair $x, y \in \underline{\mathcal{C}}$, such that the following diagrams commute for all $x, y, z \in \underline{\mathcal{C}}$:

$$\begin{tikzcd}
\underline{\mathcal{C}}(y,z) \otimes \underline{\mathcal{C}}(x,y) \arrow[r, "\circ"] \arrow[d, "F_{y,z} \otimes F_{x,y}"'] & \underline{\mathcal{C}}(x,y) \arrow[d, "F_{x,z}"] \\
\underline{\mathcal{D}}(F(y),F(z)) \otimes \underline{\mathcal{D}}(F(x),F(y)) \arrow[r, "\circ"'] & \underline{\mathcal{D}}(F(x),F(z))
\end{tikzcd}
\qquad
\begin{tikzcd}
* \arrow[r, "\mathrm{id}_x"] \arrow[dr, "\mathrm{id}_{F(x)}"'] & \underline{\mathcal{C}}(x,x) \arrow[d, "F_{x,x}"] \\
 & \underline{\mathcal{D}}(F(x),F(x)).
\end{tikzcd}$$

♡

For a regular category, hence a $\mathcal{S}$et-category, these diagrams demand that $F(- \circ -) = F(-) \circ F(-)$ and $F(1_{(-)}) = 1_{F(-)}$. More importantly, a topological functor between $\mathcal{T}$op-categories consists of continuous maps between hom-spaces. $\mathcal{T}$op-functors are frequently called *continuous* functors in the literature, an unfortunate conflict of terminology from the uncommon, but still used, notion of a continuous functor being one which preserves small limits.

A simplicial functor between s$\mathcal{S}$et-categories consists of maps between the n-simplices of the appropriate hom-spaces. A 2-functor between $\mathcal{C}$at-categories, equivalently 2-categories, is a map of objects, morphisms, and 2-cells that preserves domains, codomains, composition, and identities at all levels. Going back to a definition we explored earlier, a covariant $\mathcal{A}$b-functor from a singleton $\mathcal{A}$b-category R to $\underline{\mathcal{A}\mathrm{b}}$ is a left R-module; a contravariant $\mathcal{A}$b-functor is a right R-module.

Exercise. Extend definition 3.4 to define the underlying functor of $\mathcal{V}$-functors and show that your definition is functorial, i.e., defines a functor $(-)_0 : \mathcal{V}\text{-}\mathcal{C}\mathrm{at} \to \mathcal{C}\mathrm{at}$.

Proof. Given a $\mathcal{V}$-functor $F : \underline{\mathcal{C}} \to \underline{\mathcal{D}}$, define $F_0 : \mathcal{C}_0 \to \mathcal{D}_0$ to be the same as F on objects, and on morphisms is simply given by the relevant postcomposition in $\mathcal{V}$. More precisely, for $f : x \to y$ in $\mathcal{C}_0$, we have $f : * \to \underline{\mathcal{C}}(x, y)$ in $\mathcal{V}$, and we then define

$$F_0(f) \coloneqq * \xrightarrow{f} \underline{\mathcal{C}}(x, y) \xrightarrow{F_{x,y}} \underline{\mathcal{D}}(x, y).$$

Hence $F(f) \in \mathcal{V}(*, \underline{\mathcal{D}}(x, y)) \eqqcolon \mathcal{D}_0(x, y)$. The fact that F commutes with $\circ$ in $\mathcal{V}$ and that it preserves identities makes it functorial. These are fairly easy to check, but since this is an exercise

I will spell it out in completeness. Let $g : y \to z$ and $f : x \to y$ in $\mathcal{C}_0$. We obtain the diagram

$$\begin{tikzcd}
* \otimes * \arrow[r, "g\otimes f"] & \underline{\mathcal{C}}(y,z) \otimes \underline{\mathcal{C}}(x,y) \arrow[r, "\circ"] \arrow[d, "F_{y,z}\otimes F_{x,y}"'] & \underline{\mathcal{C}}(x,y) \arrow[d, "F_{x,z}"] \\
& \underline{\mathcal{D}}(F(y),F(z)) \otimes \underline{\mathcal{D}}(F(x),F(y)) \arrow[r, "\circ"'] & \underline{\mathcal{D}}(F(x),F(z)).
\end{tikzcd}$$

This expresses the equality

$$\circ_{\mathcal{D}} \circ (F_{y,z} \otimes F_{x,y}) \circ (g \otimes f) = F_{x,z} \circ \circ_{\mathcal{C}} \circ (g \otimes f).$$

Functorality of $\otimes$ gives on the left

$$\circ_{\mathcal{D}} \circ (F_{y,z} \otimes F_{x,y}) \circ (g \otimes f) = \circ_{\mathcal{D}} \circ ((F_{y,z} \circ g) \otimes (F_{x,y} \circ f)).$$

We thus obtain by definition of F_0 and of composition in $\mathcal{C}_0$ that

$$F_0(g) \circ F_0(f) = F_0(g \circ f).$$

The identity is manifestly preserved. Thus, F_0 is functorial.

Now for the functor $(-)_0$. We wish to show the assignment

$$\underline{\mathcal{C}} \mapsto \mathcal{C}_0, \qquad F \mapsto F_0$$

is functorial. This is fairly simple. Let $F : \underline{\mathcal{C}} \to \underline{\mathcal{D}}$ and $G : \underline{\mathcal{D}} \to \underline{\mathcal{E}}$ be $\mathcal{V}$-functors. Then by definition of functor composition, we have

$$(G \circ F)_0(f) = (G \circ F)_{x,y} \circ f = G_{x,y} \circ F_{x,y} \circ f = (G_0 \circ F_0)(f).$$

Thus the assignment is indeed functorial. ■

When $\underline{\mathcal{V}}$ is a closed $\mathcal{V}$-category, there is for any $\underline{\mathcal{C}}$ a *representable functor* $\underline{\mathcal{C}}(c,-) : \underline{\mathcal{C}} \to \underline{\mathcal{V}}$ whose underlying functor is the representable functor $\underline{\mathcal{C}}(c,-) : \mathcal{C}_0 \to \mathcal{V}$ as constructed earlier. The object map is merely $x \mapsto \underline{\mathcal{C}}(c,x)$; the morphisms

$$\underline{\mathcal{C}}(c,-)_{x,y} : \underline{\mathcal{C}}(x,y) \to \underline{\mathcal{V}}(\underline{\mathcal{C}}(c,x), \underline{\mathcal{C}}(c,y))$$

are the adjuncts of composition. More precisely, since $\underline{\mathcal{V}}$ is closed, composition

$$\circ : \underline{\mathcal{C}}(x,y) \otimes \underline{\mathcal{C}}(c,x) \to \underline{\mathcal{C}}(c,y)$$

has an adjunct under definition 3.2 to

$$\underline{\mathcal{C}}(x,y) \to \underline{\mathcal{V}}(\underline{\mathcal{C}}(c,x), \underline{\mathcal{C}}(c,y)).$$

The definition of composition in the underlying category recovers $g \mapsto g_*$ for the underlying functor.

Suppose still that $\mathcal{V}$ is closed monoidal. Recall an unenriched category $\mathcal{C}$ admits a free $\mathcal{V}$-categorical structure with objects the same as $\mathcal{C}$, and hom-objects given by

$$\underline{\mathcal{C}}(a,b) \coloneqq \mathcal{C}(a,b) \cdot * = \coprod_{\mathcal{C}(a,b)} *.$$

A functor F from the free $\mathcal{V}$-category on $\mathcal{C}$ to a $\mathcal{V}$-category $\underline{\mathcal{D}}$ consists of an object function $c \mapsto F(c)$, together with morphisms

$$F_{x,y} : \coprod_{\mathcal{C}(x,y)} * \to \underline{\mathcal{D}}(F(x), F(y)).$$

Universality of coproduct gives that $F_{x,y}$ is completely determined by some family of morphisms $X \coloneqq \{\varphi_f : * \to \underline{\mathcal{D}}(F(x), F(y))\}_{f \in \mathcal{C}(x,y)}$. Since

$$X \subseteq \mathcal{V}(*, \underline{\mathcal{D}}(F(x), F(y))) \eqqcolon \mathcal{D}_0(F(x), F(y)),,$$

we have that $F_{x,y}$ is equivalently an assignment $\mathcal{C}(x,y) \to \mathcal{D}_0(F(x), F(y))$. Assigning the notation $F^V(\mathcal{C})$ to the free $\mathcal{V}$-category on $\mathcal{C}$ (this notation will never be used again and is simply here to write down precisely my point), we have that functors $F : F^V(\mathcal{C}) \to \underline{\mathcal{D}}$ are precisely functors $F : \mathcal{C} \to \mathcal{D}_0$. Hence, there is an isomorphism

$$\mathcal{V}\text{-}\mathcal{C}\mathrm{at}(F^V(\mathcal{C}), \underline{\mathcal{D}}) \cong \mathcal{C}\mathrm{at}(\mathcal{C}, \mathcal{D}_0)$$

natural in both entries by universality of coproduct. Thus there is an adjunction $F^V \dashv (-)_0$, and for this reason we will never again use the notation F^V, since functors $F^V(\mathcal{C}) \to \underline{\mathcal{D}}$ are simply functors $\mathcal{C} \to \mathcal{D}_0$.

Recall that $\mathrm{s}\mathcal{S}\mathrm{et}$ admits a closed enrichment via $\underline{\mathrm{s}\mathcal{S}\mathrm{et}}(X,Y)_n = \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n \times X, Y)$. The terminal simplicial set $\Delta^0 = \mathbf{\Delta}(-, [0])$ in an unenriched sense also represents the functor $(-) : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{S}\mathrm{et}$, which takes a simplicial set X to it's set X_0 of vertices. However viewing $\mathrm{s}\mathcal{S}\mathrm{et}$ as self-enriched, the represented functor $\underline{\mathrm{s}\mathcal{S}\mathrm{et}}(\Delta^0, -)$ becomes the identity on $\mathrm{s}\mathcal{S}\mathrm{et}$:

$$\underline{\mathrm{s}\mathcal{S}\mathrm{et}}(\Delta^0, X)_n = \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n \times \Delta^0, X) \cong \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n, X) \cong X_n.$$

The equality is definition, the first isomorphism is that Δ^0 is terminal, and the second is Yoneda.

The next natural definition is that of a $\mathcal{V}$-natural transformation. However, we need to recall that there are no morphisms to act on, so the natural way to reproduce the naturality square

$$\begin{tikzcd} F(x) \arrow[r, "F(f)"] \arrow[d, "\alpha_x"'] & F(y) \arrow[d, "\alpha_y"] \\ G(x) \arrow[r, "G(f)"'] & G(y) \end{tikzcd}$$

is via pre- and post-composition with components of α.

Definition 3.6 ($\mathcal{V}$-Natural Transformation)**.** A $\mathcal{V}$-natural transformation $\alpha : F \Rightarrow G$ between $\mathcal{V}$-functors $F, G : \underline{\mathcal{C}} \to \underline{\mathcal{D}}$ is a morphism $\alpha_x : * \to \underline{\mathcal{D}}(F(x), G(x))$ in $\mathcal{V}$ for each $x \in \underline{\mathcal{C}}$, such that for all $x, y \in \underline{\mathcal{C}}$, the diagram

$$\begin{tikzcd} \underline{\mathcal{C}}(x,y) \arrow[r, "F_{x,y}"] \arrow[d, "G_{x,y}"'] & \underline{\mathcal{D}}(F(x), F(y)) \arrow[d, "(\alpha_y)_*"] \\ \underline{\mathcal{D}}(G(x), G(y)) \arrow[r, "(\alpha_x)^*"'] & \underline{\mathcal{D}}(F(x), G(y)) \end{tikzcd}$$

with precomposition $(\alpha_x)^*$ and postcomposition $(\alpha_y)_*$ defined as earlier. ♡

The intuitive understanding of this in unenriched categories is that on the top right path, morphisms $f : x \to y$ in $\mathcal{C}$ should be carried to morphisms $F(f) : F(x) \to F(y)$ in $\mathcal{D}$, and then postcomposed with α_y. The bottom left path says that this shouldbe equal to taking your morphism and applying G to obtain $G(f) : G(x) \to G(y)$, then precompositing with α_x, giving an equality

$$\alpha_y \circ F(f) = G(f) \circ \alpha_x.$$

This is precisely the naturality condition.

Example. An $\mathcal{A}$b-natural transformation between two left R-modules $A, B : R \to \mathcal{A}\mathrm{b}$ is a single arrow $\alpha_* : * \to \mathcal{A}\mathrm{b}(A, B)$ in $\mathcal{A}$b, and hence a group homomorphism, that must commute with scalar multiplication by our above discussion. Hence $\mathcal{A}$b-natural transformations here are R-module homomorphisms, and thus $\mathcal{A}\mathrm{b}\text{-}\operatorname{Fun}(R, \mathcal{A}\mathrm{b}) \cong \mathcal{M}\mathrm{od}_R$.

The addition of $\mathcal{V}$-natural transformations assembles $\mathcal{V}$-$\mathcal{C}$at into a 2-category. We've shown already that the underlying set functor $(-)_0$ extends to $\mathcal{V}$-functors, making it a functor $\mathcal{V}$-$\mathcal{C}\mathrm{at} \to \mathcal{C}\mathrm{at}$. We now claim we can use it to define underlying $\mathcal{V}$-natural transformations.

Proposition 3.1. *Underlying $(-)_0$ is a 2-functor $\mathcal{V}$-$\mathcal{C}\mathrm{at} \to \mathcal{C}\mathrm{at}$. Additionally, lax monoidal functors $F : \mathcal{V} \to \mathcal{U}$ induce 2-functors $\mathcal{V}$-$\mathcal{C}\mathrm{at} \to \mathcal{U}$-$\mathcal{C}\mathrm{at}$.*

Proof. The statement is obvious, yet the proof is messy. Let $(\mathcal{V}, \otimes, *)$ and $(\mathcal{U}, \otimes, 1)$ be monoidal, and let $F : \mathcal{V} \to \mathcal{U}$ be lax monoidal. Let $\lambda : F(x) \otimes F(x) \to F(x \otimes x)$ be the natural transformation, and let $\varphi : 1 \to F(*)$ be the other arrow.

We prove the second statement first. By lemma 3.6, we already have an object function by mapping some $\mathcal{V}$-category $\underline{\mathcal{D}}_{\mathcal{V}}$ to the $\mathcal{U}$-category $\underline{\mathcal{D}}_{\mathcal{U}}$ with the same objects, and hom-objects

$$\underline{\mathcal{D}}_{\mathcal{U}}(x, y) \coloneqq F\left(\underline{\mathcal{D}}_{\mathcal{V}}(x, y)\right).$$

Let $G : \underline{\mathcal{C}}_{\mathcal{V}} \to \underline{\mathcal{D}}_{\mathcal{V}}$ be a $\mathcal{V}$-functor. Define a new functor $F(G)$ by $F(G) = G$ on objects, and

$$(F(G))_{x,y} = F(G_{x,y}).$$

Since G is $\mathcal{V}$-functorial, the diagram

$$\begin{tikzcd}
\underline{\mathcal{C}}_{\mathcal{V}}(y,z) \otimes \underline{\mathcal{C}}_{\mathcal{V}}(x,y) \arrow[r, "\circ"] \arrow[d, "G_{y,z} \otimes G_{x,y}"'] & \underline{\mathcal{C}}_{\mathcal{V}}(x,z) \arrow[d, "G_{x,z}"] \\
\underline{\mathcal{D}}_{\mathcal{V}}(y,z) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(x,y) \arrow[r, "\circ"'] & \underline{\mathcal{D}}_{\mathcal{V}}(x,z)
\end{tikzcd}$$

commutes in $\mathcal{V}$. Hence by functorality of F, the diagram

$$\begin{tikzcd}
F(\underline{\mathcal{C}}_{\mathcal{V}}(y,z) \otimes \underline{\mathcal{C}}_{\mathcal{V}}(x,y)) \arrow[r, "F(\circ)"] \arrow[d, "F(G_{y,z} \otimes G_{x,y})"'] & \underline{\mathcal{C}}_{\mathcal{U}}(x,z) \arrow[d, "F(G_{x,z})"] \\
F(\underline{\mathcal{D}}_{\mathcal{V}}(y,z) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(x,y)) \arrow[r, "F(\circ)"'] & \underline{\mathcal{D}}_{\mathcal{U}}(x,z)
\end{tikzcd}$$

commutes in $\mathcal{U}$, where we have swapped for the $\mathcal{U}$-categories where possible. Recall since F is lax monoidal, there is a natural transformation

$$F(v) \otimes F(v') \to F(v \otimes v').$$

More precisely, there is a natural transformation

$$F(-) \otimes F(-) \Rightarrow F(- \otimes -).$$

Denote this natural transformation by λ. Since precomposing in the corner preserves commutivity, the diagram

$$\begin{tikzcd}
\underline{\mathcal{C}}_{\mathcal{U}}(y,z) \otimes \underline{\mathcal{C}}_{\mathcal{U}}(x,y) \arrow[r, "\lambda"] & F(\underline{\mathcal{C}}_{\mathcal{V}}(y,z) \otimes \underline{\mathcal{C}}_{\mathcal{V}}(x,y)) \arrow[r, "F(\circ)"] \arrow[d, "F(G_{y,z} \otimes G_{x,y})"'] & \underline{\mathcal{C}}_{\mathcal{U}}(x,z) \arrow[d, "F(G_{x,z})"] \\
& F(\underline{\mathcal{D}}_{\mathcal{V}}(y,z) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(x,y)) \arrow[r, "F(\circ)"'] & \underline{\mathcal{D}}_{\mathcal{U}}(x,z)
\end{tikzcd}$$

commutes in $\mathcal{U}$. Thus, by naturality of λ, the diagram

$$\begin{tikzcd}
\underline{\mathcal{C}}_{\mathcal{U}}(y,z) \otimes \underline{\mathcal{C}}_{\mathcal{U}}(x,y) \arrow[r, "\lambda"] \arrow[d, "F(G_{y,z}) \otimes F(G_{x,y})"'] & F(\underline{\mathcal{C}}_{\mathcal{V}}(y,z) \otimes \underline{\mathcal{C}}_{\mathcal{V}}(x,y)) \arrow[r, "F(\circ)"] \arrow[d, "F(G_{y,z} \otimes G_{x,y})"'] & \underline{\mathcal{C}}_{\mathcal{U}}(x,z) \arrow[d, "F(G_{x,z})"] \\
\underline{\mathcal{D}}_{\mathcal{U}}(y,z) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(x,y) \arrow[r, "\lambda"'] & F(\underline{\mathcal{D}}_{\mathcal{V}}(y,z) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(x,y)) \arrow[r, "F(\circ)"'] & \underline{\mathcal{D}}_{\mathcal{U}}(x,z)
\end{tikzcd}$$

commutes as well. Recall by lemma 3.6 that composition is given in the $\mathcal{U}$-categories via

$$\underline{\mathcal{C}}_{\mathcal{U}}(y,z) \otimes \underline{\mathcal{C}}_{\mathcal{U}}(x,y) \xrightarrow{\lambda} F\left(\underline{\mathcal{C}}_{\mathcal{V}}(y,z) \otimes \underline{\mathcal{C}}_{\mathcal{V}}(x,y)\right) \xrightarrow{F(\circ)} \underline{\mathcal{C}}_{\mathcal{U}}(x,z).$$

Note that this is the top and bottom compositions of the above diagram, so simplifying we obtain the diagram

$$\begin{tikzcd}
\underline{\mathcal{C}}_{\mathcal{U}}(y,z) \otimes \underline{\mathcal{C}}_{\mathcal{U}}(x,y) \arrow[r, "\circ"] \arrow[d, "F(G_{y,z}) \otimes F(G_{x,y})"'] & \underline{\mathcal{C}}_{\mathcal{U}}(x,z) \arrow[d, "F(G_{x,z})"] \\
\underline{\mathcal{D}}_{\mathcal{U}}(y,z) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(x,y) \arrow[r, "\circ"'] & \underline{\mathcal{D}}_{\mathcal{U}}(x,z).
\end{tikzcd}$$

Thus, $F(G)$ preserves composition.

To show $F(G)$ preserves identities, recall by definition 3.5 that since G is $\mathcal{V}$-functorial, for any x the diagram

$$\begin{tikzcd} * \arrow[r, "\mathrm{id}_x"] \arrow[dr, "\mathrm{id}_{G(x)}"'] & \underline{\mathcal{C}}_{\mathcal{V}}(x,x) \arrow[d, "G_{x,x}"] \\ & \underline{\mathcal{D}}_{\mathcal{V}}(G(x),G(x)) \end{tikzcd}$$

commutes. Functorality of F gives that the diagram

$$\begin{tikzcd} F(*) \arrow[r, "F(\mathrm{id}_x)"] \arrow[dr, "F(\mathrm{id}_{G(x)})"'] & \underline{\mathcal{C}}_{\mathcal{U}}(x,x) \arrow[d, "F(G_{x,x})"] \\ & \underline{\mathcal{D}}_{\mathcal{U}}(G(x),G(x)) \end{tikzcd}$$

commutes. To avoid confusion, denote by 1 the unit in $\mathcal{U}$. Since F is lax monoidal, there is a natural transformation $1 \to F(*)$, giving that

$$\begin{tikzcd} 1 \arrow[r] & F(*) \arrow[r, "F(\mathrm{id}_x)"] \arrow[dr, "F(\mathrm{id}_{G(x)})"'] & \underline{\mathcal{C}}_{\mathcal{U}}(x,x) \arrow[d, "F(G_{x,x})"] \\ & & \underline{\mathcal{D}}_{\mathcal{U}}(G(x),G(x)) \end{tikzcd}$$

commutes. Recall also that lemma 3.6 defines the identity as the composition

$$\mathrm{id}_x : 1 \longrightarrow F(*) \xrightarrow{F(\mathrm{id}_x)} \underline{\mathcal{D}}_{\mathcal{U}}(x,x).$$

Note this is the top composition on the above diagram, and also note that we have by the same result that

$$\mathrm{id}_{G(x)} : 1 \longrightarrow F(*) \xrightarrow{F(\mathrm{id}_{G(x)})} \underline{\mathcal{D}}_{\mathcal{U}}(G(x),G(x)).$$

Note that this is the other side of the diagram. Thus, the diagram is equivalent to

$$\begin{tikzcd} 1 \arrow[r, "\mathrm{id}_x"] \arrow[dr, "\mathrm{id}_{G(x)}"'] & \underline{\mathcal{C}}_{\mathcal{U}}(x,x) \arrow[d, "F(G_{x,x})"] \\ & \underline{\mathcal{D}}_{\mathcal{U}}(G(x),G(x)). \end{tikzcd}$$

Thus, $F(G)$ is a $\mathcal{U}$-functor, and F is a functor.

Finally, we must show F is a 2-functor by showing it preserves domain, codomain, identity, and composition of natural transformations both horizontally and vertically. Let $G, H : \underline{\mathcal{C}}_{\mathcal{V}} \to \underline{\mathcal{D}}_{\mathcal{V}}$ be $\mathcal{V}$-functors, and let $\alpha : G \Rightarrow H$ be a $\mathcal{V}$-natural transformation. Let $\varphi : 1 \to F(*)$ be the relevant lax monoidal arrow. Since the components α_x are morphisms in $\mathcal{V}$, define $F(\alpha)$ componentwise as $F(\alpha)_x = F(\alpha_x) \circ \varphi$. This is indeed an arrow $1 \to \underline{\mathcal{D}}_{\mathcal{U}}(G(x), H(x))$ by definition of $\underline{\mathcal{D}}_{\mathcal{U}}$. By definition 3.6, the square

$$\begin{tikzcd}
\underline{\mathcal{C}}_{\mathcal{V}}(x,y) \arrow[r, "G_{x,y}"] \arrow[d, "H_{x,y}"'] & \underline{\mathcal{D}}_{\mathcal{V}}(G(x),G(y)) \arrow[d, "(\alpha_y)_*"] \\
\underline{\mathcal{D}}_{\mathcal{V}}(H(x),H(y)) \arrow[r, "(\alpha_x)^*"'] & \underline{\mathcal{D}}_{\mathcal{V}}(G(x),H(y))
\end{tikzcd}$$

commutes. Functorality of F gives that the square

$$\begin{tikzcd}
\underline{\mathcal{C}}_{\mathcal{U}}(x,y) \arrow[r, "F(G_{x,y})"] \arrow[d, "F(H_{x,y})"'] & \underline{\mathcal{D}}_{\mathcal{U}}(G(x),G(y)) \arrow[d, "F((\alpha_y)_*)"] \\
\underline{\mathcal{D}}_{\mathcal{U}}(H(x),H(y)) \arrow[r, "F((\alpha_x)^*)"'] & \underline{\mathcal{D}}_{\mathcal{U}}(G(x),H(y))
\end{tikzcd}$$

commutes. Recall that postcomposition is given by the composition

$$(\alpha_y)_* : * \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x),G(y)) \xrightarrow{\alpha_y \otimes 1} \underline{\mathcal{D}}_{\mathcal{V}}(G(x),H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x),G(y)) \xrightarrow{\circ} \underline{\mathcal{D}}_{\mathcal{V}}(G(x),H(y)).$$

Applying F and naturality of the lax functorial arrow λ gives the diagram

$$\begin{tikzcd}
F(* \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x),G(y))) \arrow[r, "F(\alpha_y \otimes 1)"] & F(\underline{\mathcal{D}}_{\mathcal{V}}(G(y),H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x),G(y))) \arrow[r, "F(\circ)"] & \underline{\mathcal{D}}_{\mathcal{U}}(G(x),H(y)) \\
F(*) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x),G(y)) \arrow[u, "\lambda"] \arrow[r, "F(\alpha_y)\otimes 1"'] & \underline{\mathcal{D}}_{\mathcal{U}}(G(y),H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x),G(y)) \arrow[u, "\lambda"'] &
\end{tikzcd}$$

commuting. Applying this to the side of the above diagram gives

$$\begin{tikzcd}
\underline{\mathcal{C}}_{\mathcal{U}}(x,y) \arrow[r, "F(G_{x,y})"] \arrow[ddd, "F(H_{x,y})"'] & \underline{\mathcal{D}}_{\mathcal{U}}(G(x),G(y)) \arrow[d, "F(\ell_{\mathcal{V}}^{-1})"] & \\
& F\left(* \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x),G(y))\right) \arrow[d, "F(\alpha_y \otimes 1)"] & F(*) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x),G(y)) \arrow[l, "\lambda"'] \arrow[d, "F(\alpha_y)\otimes 1"] \\
& F\left(\underline{\mathcal{D}}_{\mathcal{V}}(G(y),H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x),G(y))\right) \arrow[d, "F(\circ)"] & \underline{\mathcal{D}}_{\mathcal{U}}(G(y),H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x),G(y)) \arrow[l, "\lambda"'] \\
\underline{\mathcal{D}}_{\mathcal{U}}(H(x),H(y)) \arrow[r, "F((\alpha_x)^*)"'] & \underline{\mathcal{D}}_{\mathcal{U}}(G(x),H(y)) &
\end{tikzcd}$$

which commutes for $\ell_{\mathcal{V}}$ the left unitor in $\mathcal{V}$. One of the coherence conditions for lax monoidal functors states that, for $\varphi : 1 \to F(*)$ the unit arrow for the lax monoidal functor and for $\ell_{\mathcal{U}}$ the left unitor in $\mathcal{U}$, for any $x \in \mathcal{V}$, the diagram

$$\begin{tikzcd}
1 \otimes F(x) \arrow[r, "\varphi\otimes 1"] \arrow[d, "\ell_{\mathcal{U}}"'] & F(*) \otimes F(x) \arrow[d, "\lambda"] \\
F(x) & F(* \otimes x) \arrow[l, "F(\ell_{\mathcal{V}})"]
\end{tikzcd}$$

commutes. Postcomposing with $F(\ell_{\mathcal{V}}^{-1}) = F(\ell_{\mathcal{V}})^{-1}$ and precomposing with $\ell_{\mathcal{U}}^{-1}$ give the equalities

$$F(\ell_{\mathcal{V}}^{-1}) \circ \ell_{\mathcal{U}} = \lambda \circ (\varphi \otimes 1) \implies F(\ell_{\mathcal{V}}^{-1}) = \lambda \circ (\varphi \otimes 1) \circ \ell_{\mathcal{U}}^{-1}.$$

Plugging in $x = \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y))$ and applying this equality gives the commutative square

$$\begin{tikzcd}
1 \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \arrow[r, "\varphi\otimes 1"] & F(*) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \arrow[d, "\lambda"] \\
\underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \arrow[u, "\ell_{\mathcal{U}}^{-1}"] \arrow[r, "F(\ell_{\mathcal{V}}^{-1})"'] & F(* \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y))).
\end{tikzcd}$$

Rotating 90 degrees clockwise allows us to plug this into the top right corner of the above diagram, giving the commutative diagram (left side omitted for simplicity)

$$\begin{tikzcd}
\underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \arrow[r, "\ell_{\mathcal{U}}^{-1}"] \arrow[d, "F(\ell_{\mathcal{V}}^{-1})"] & 1 \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \arrow[d, "\varphi\otimes 1"] \\
F\left(* \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y))\right) \arrow[d, "F(\alpha_y \otimes 1)"] & F(*) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \arrow[l, "\lambda"] \arrow[d, "F(\alpha_y)\otimes 1"] \\
F\left(\underline{\mathcal{D}}_{\mathcal{V}}(G(y), H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(G(x), G(y))\right) \arrow[d, "F(\circ)"] & \underline{\mathcal{D}}_{\mathcal{U}}(G(y), H(y)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(G(x), G(y)) \arrow[l, "\lambda"] \\
\underline{\mathcal{D}}_{\mathcal{U}}(G(x), H(y)). &
\end{tikzcd}$$

This finally gives an expression for $F((\alpha_y)_*)$ as the composition (here recognizing that $\circ := F(\circ) \circ \lambda$ in $\underline{\mathcal{D}}_{\mathcal{U}}$)

$$F((\alpha_y)_*) = F(\circ) \circ \lambda \circ F(\alpha_y) \otimes 1 \circ (\varphi \otimes 1) \circ \ell_{\mathcal{U}}^{-1}.$$

This is precisely the definition of $(F(\alpha_y) \circ \varphi)_* = (F(\alpha_y))_*$. Verification of precomposition is the exact same, since φ moves with the morphism to the other side of the monoidal product. Therefore, $F(\alpha) : F(G) \to F(H)$ is $\mathcal{U}$-natural. In particular, domain and codomain are preserved.

Note that the identity $\mathcal{V}$-natural transformation $1_G : G \Rightarrow G$ is obviously given by $(1_G)_x = \mathrm{id}^{\mathcal{V}}_{G(x)}$. By definition, $F(1_G)_x = F(\mathrm{id}^{\mathcal{V}}_{G(x)}) \circ \varphi$. Recall that we have by definition

$$F(\mathrm{id}^{\mathcal{V}}_{G(x)}) \circ \varphi = \mathrm{id}^{\mathcal{U}}_{G(x)}.$$

Moreover, $F(G(x)) = G(x)$, and thus $(1_{F(G)})_x = \mathrm{id}_{G(x)}$. Therefore, $F(1_G) = 1_{F(G)}$ preserves identities.

It thus suffices to show F preserves horizontal and vertical composition. First, we show vertical composition. Let $G, H, I : \underline{\mathcal{C}}_{\mathcal{V}} \to \underline{\mathcal{D}}_{\mathcal{V}}$ be $\mathcal{V}$-functors, and let $\alpha : G \Rightarrow H$ and $\beta : H \Rightarrow I$ be $\mathcal{V}$-natural transformations. Define the horizontal composite $(\beta \circ \alpha)$ componentwise by composing in the underlying category as

$$(\beta \circ \alpha)_x : * \xrightarrow{\ell_{\mathcal{V}}^{-1}} * \otimes * \xrightarrow{\beta_x \otimes \alpha_x} \underline{\mathcal{D}}_{\mathcal{V}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(H(x), G(x)) \xrightarrow{\circ} \underline{\mathcal{D}}_{\mathcal{V}}(I(x), G(x)).$$

It suffices to show $F(\beta \circ \alpha)_x = F(\beta)_x \circ F(\alpha)_x$. We defined

$$F(\beta \circ \alpha)_x = F((\beta \circ \alpha)_x) \circ \varphi.$$

This is viewed by functoriality of F on the above composition as

$$1 \xrightarrow{\varphi} F(*) \xrightarrow{F(\ell_{\mathcal{V}}^{-1})} F(* \otimes *) \xrightarrow{F(\beta_x \otimes \alpha_x)} F\left(\underline{\mathcal{D}}_{\mathcal{V}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(H(x), G(x))\right) \xrightarrow{F(\circ)} \underline{\mathcal{D}}_{\mathcal{U}}(I(x), G(x)).$$

By naturality of λ, it follows that

$$\begin{tikzcd}
 & F(*) \otimes F(*) \arrow[r, "F(\beta_x) \otimes F(\alpha_x)"] \arrow[d, "\lambda"'] & \underline{\mathcal{D}}_{\mathcal{U}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(H(x), G(x)) \arrow[d, "\lambda"] & \\
1 \arrow[r, "\varphi"'] & F(*) \arrow[r, "F(\ell_{\mathcal{V}}^{-1})"'] & F(* \otimes *) \arrow[r, "F(\beta_x \otimes \alpha_x)"'] & F\left(\underline{\mathcal{D}}_{\mathcal{V}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(H(x), G(x))\right) \arrow[r, "F(\circ)"'] & \underline{\mathcal{D}}_{\mathcal{U}}(I(x), G(x))
\end{tikzcd}$$

commutes. Applying the coherence condition for the left unitor as we did before, we obtain that the diagram

$$\begin{tikzcd}
 & 1 \otimes F(*) \arrow[r, "\varphi \otimes 1"] & F(*) \otimes F(*) \arrow[r, "F(\beta_x) \otimes F(\alpha_x)"] \arrow[d, "\lambda"'] & \underline{\mathcal{D}}_{\mathcal{U}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(H(x), G(x)) \arrow[d, "\lambda"] & \\
1 \arrow[r, "\varphi"'] & F(*) \arrow[u, "\ell_{\mathcal{U}}^{-1}"] \arrow[r, "F(\ell_{\mathcal{V}}^{-1})"'] & F(* \otimes *) \arrow[r, "F(\beta_x \otimes \alpha_x)"'] & F\left(\underline{\mathcal{D}}_{\mathcal{V}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(H(x), G(x))\right) \arrow[r, "F(\circ)"'] & \underline{\mathcal{D}}_{\mathcal{U}}(I(x), G(x))
\end{tikzcd}$$

also commutes. Note that since $\ell_{\mathcal{U}} : * \otimes - \Rightarrow 1_{\mathcal{U}}$, is natural, we also have that $\ell_{\mathcal{U}}^{-1} : 1 \Rightarrow * \otimes -$ is natural. Hence by naturality of $\ell_{\mathcal{U}}^{-1}$,

$$\begin{tikzcd}
1 \arrow[r, "\varphi"] \arrow[d, "\ell_{\mathcal{U}}^{-1}"'] & F(*) \arrow[d, "\ell_{\mathcal{U}}^{-1}"] \\
1 \otimes 1 \arrow[r, "1\otimes\varphi"'] & 1 \otimes F(*)
\end{tikzcd}$$

commutes. Gluing this into the top left of our above diagram gives the commuative diagram gives us that the diagram

$$\begin{tikzcd}
1 \otimes 1 \arrow[r, "1\otimes\varphi"] & 1 \otimes F(*) \arrow[r, "\varphi\otimes 1"] & F(*) \otimes F(*) \arrow[r, "F(\beta_x)\otimes F(\alpha_x)"] \arrow[d, "\lambda"'] & \underline{\mathcal{D}}_{\mathcal{U}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(H(x), G(x)) \arrow[d, "\lambda"] & \\
1 \arrow[u, "\ell_{\mathcal{U}}^{-1}"] \arrow[r, "\varphi"'] & F(*) \arrow[u, "\ell_{\mathcal{U}}^{-1}"'] \arrow[r, "F(\ell_{\mathcal{V}}^{-1})"'] & F(* \otimes *) \arrow[r, "F(\beta_x\otimes\alpha_x)"'] & F\left(\underline{\mathcal{D}}_{\mathcal{V}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{V}}(H(x), G(x))\right) \arrow[r, "F(\circ)"'] & \underline{\mathcal{D}}_{\mathcal{U}}(I(x), G(x))
\end{tikzcd}$$

commutes. Note once again that as defined in lemma 3.6, $\lambda \circ F(\circ)$ is simply composition in $\underline{\mathcal{D}}_{\mathcal{U}}$, and since $(\varphi \otimes 1) \circ (1 \otimes \varphi) = \varphi \otimes \varphi$ by functorality, this gives us the equality

$$\begin{tikzcd}
1 \arrow[r, "\ell_{\mathcal{U}}^{-1}"'] \arrow[rrrr, bend left, "F(\beta\circ\alpha)_x"] & F(*) \otimes F(*) \arrow[r, "\varphi\otimes\varphi"'] & F(*) \otimes F(*) \arrow[r, "F(\beta_x)\otimes F(\alpha_x)"'] & \underline{\mathcal{D}}_{\mathcal{U}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(H(x), G(x)) \arrow[r, "\circ"'] & \underline{\mathcal{D}}_{\mathcal{U}}(I(x), G(x)),
\end{tikzcd}$$

where we have recalled that the bottom row above is the expanded version of $F(\beta \circ \alpha)_x = F(\beta_x \circ \alpha_x) \circ \varphi$. Functorality of $\otimes$ once again gives that

$$(F(\beta_x) \otimes F(\alpha_x)) \circ (\varphi \otimes \varphi) = (F(\beta_x) \circ \varphi) \otimes (F(\alpha_x) \circ \varphi) = F(\beta)_x \otimes F(\alpha)_x.$$

Applying this to the above diagram gives

$$\begin{tikzcd}
1 \arrow[r, "\ell_{\mathcal{U}}^{-1}"'] \arrow[rrr, bend left, "F(\beta\circ\alpha)_x"] & F(*) \otimes F(*) \arrow[r, "F(\beta)_x\otimes F(\alpha)_x"'] & \underline{\mathcal{D}}_{\mathcal{U}}(I(x), H(x)) \otimes \underline{\mathcal{D}}_{\mathcal{U}}(H(x), G(x)) \arrow[r, "\circ"'] & \underline{\mathcal{D}}_{\mathcal{U}}(I(x), G(x)).
\end{tikzcd}$$

Note that this is precisely the definition of composition $F(\beta)_x \circ F(\alpha)_x$ in $\underline{\mathcal{D}}_{\mathcal{U}}$. Thus, the diagram expresses the equaltiy

$$F(\beta \circ \alpha)_x = F(\beta)_x \circ F(\alpha)_x.$$

Therefore, F preserves vertical composition of $\mathcal{V}$-natural transformastions.

Finally, let

$$\begin{tikzcd} \underline{\mathcal{C}}_{\mathcal{V}} \arrow[r, bend left, "G^1"] \arrow[r, bend right, "H^1"'] \arrow[r, phantom, "\Downarrow\alpha"] & \underline{\mathcal{D}}_{\mathcal{V}} \arrow[r, bend left, "G^2"] \arrow[r, bend right, "H^2"'] \arrow[r, phantom, "\Downarrow\beta"] & \underline{\mathcal{E}}_{\mathcal{V}} \end{tikzcd}$$

be a diagram of $\mathcal{V}$-natural transformations between $\mathcal{V}$-functors. Recall the horizontal composite $\beta \cdot \alpha$ is defined componentwise as

$$(\beta \cdot \alpha)_x = (\beta_{H^1(x)}) \circ \left(G^2_{G^1(x),H^1(x)} \circ \alpha_x \right)$$

where the center composition $\circ$ is componentwise vertical composition of $\mathcal{V}$-natural transformations, and the right composition is composition in $\mathcal{V}$. We need to show $F(\beta \cdot \alpha) = F(\beta) \cdot F(\alpha)$. By definition,

$$F(\beta \cdot \alpha)_x = F\left(\beta_{H^1(x)} \circ \left(G^2_{G^1(x),H^1(x)} \circ \alpha_x \right) \right) \circ \varphi.$$

Since F preserves vertical composition, we have

$$F\left(\beta_{H^1(x)} \circ \left(G^2_{G^1(x),H^1(x)} \circ \alpha_x \right) \right) \circ \varphi = F(\beta_{H^1(x)}) \circ F\left(G^2_{G^1(x),H^1(x)} \circ \alpha_x \right) \circ \varphi.$$

Expanding out $F(\beta)_x \cdot F(\alpha)_x$ similarly gives

$$\begin{aligned} F(\beta_x) \circ F(\alpha)_x &= F(\beta)_{F(H^1)(x)} \circ \left(F\left(G^2\right)_{G^1(x),H^1(x)} \circ F(\alpha)_x \right) \\ &= F(\beta)_{H^1(x)} \circ \left(F\left(G^2_{G^1(x),H^1(x)} \right) \circ F(\alpha)_x \right) \\ &= (F(\beta_{H^1(x)}) \circ \varphi) \circ \left(F\left(G^2_{G^1(x),H^1(x)} \right) \circ F(\alpha_x) \circ \varphi \right). \end{aligned}$$

At this point the diagrams become to large to fit on the screen, so we trust the reader to use the definition of the horizonal composite (and the equalities induced when F preserves it) and to reference the large diagram used in the proof of the vertical composite being preserved, in order to see that these two equations are indeed equal. Therefore, $F : \mathcal{V}\text{-}\mathcal{C}\mathrm{at} \to \mathcal{U}\text{-}\mathcal{C}\mathrm{at}$ is a 2-functor.

Note that $(-)_0$ is simply the functor $\mathcal{V}(*, -) : \mathcal{V} \to \mathcal{S}\mathrm{et}$. Since $\mathcal{S}\mathrm{et}\text{-}\mathcal{C}\mathrm{at} = \mathcal{C}\mathrm{at}$ by definition, it suffices to show $(-)_0$ is lax monoidal with $\mathcal{S}\mathrm{et}$ endowed with the cartesian monoidal structure. First, let $\lambda : \mathcal{V}(*, -) \times \mathcal{V}(*, -) \Rightarrow \mathcal{V}(*, - \otimes -)$ be given by tensoring and precomposing with $\ell_{\mathcal{V}}^{-1}$:

$$(u, v) \mapsto (u \otimes v) \circ \ell_{\mathcal{V}}^{-1}.$$

Note that this is indeed well-defined, since $u : * \to a$ and $v : * \to b$ means that we obtain the composition

$$* \xrightarrow{\ell_{\mathcal{V}}^{-1}} * \otimes * \xrightarrow{u \otimes v} a \otimes b.$$

Now we show λ is natural. That is, for any pair (f, g) with $f : a \to c$ and $g : b \to d$ in $\mathcal{V}$, the diagram

$$\begin{tikzcd}
\mathcal{V}(*, a) \times \mathcal{V}(*, b) \arrow[r, "f_* \times g_*"] \arrow[d, "\lambda_{a,b}"'] & \mathcal{V}(*, c) \times \mathcal{V}(*, d) \arrow[d, "\lambda_{c,d}"] \\
\mathcal{V}(*, a \otimes b) \arrow[r, "(f \otimes g)_*"'] & \mathcal{V}(*, c \otimes d)
\end{tikzcd}$$

should commute. Indeed, the diagram expresses the equality

$$(f \otimes g) \circ (u \otimes v) \circ \ell_{\mathcal{V}}^{-1} = ((f \circ u) \otimes (g \circ v)) \circ \ell_{\mathcal{V}}^{-1},$$

which follows simply by functoriality of the monoidal product. Next, we need to show there is an arrow $1 \to \mathcal{V}(*, *)$, where 1 denotes the singleton. To choose this canonically, let $\varphi = \mathrm{id}_*$. Thus, a canonical arrow exists, and therefore, $(-)_{)} = \mathcal{V}(*, -)$ is a 2-functor $\mathcal{V}\text{-}\mathcal{C}\mathrm{at} \to \mathcal{C}\mathrm{at}$. ■

We now modify our notation by dropping the subscript 0. For a $\mathcal{V}$-category $\underline{\mathcal{C}}$, we denote by $\mathcal{C}$ the underlying category, and other underlying structures may be inferred from context. We also obtain from this proof an explicit description of the notion of the underlying natural transformation. Let $\underline{\alpha} : F \Rightarrow G$ be $\mathcal{V}$-natural, and define the underlying natural transformation α componentwise via

$$\alpha_x = \mathcal{V}(*, \underline{\alpha_x}) \circ \mathrm{id}_* = (\alpha_x)_*.$$

Lemma 3.4. *Let $\mathcal{V}$ be closed, and let $\underline{\mathcal{C}}$ be a $\mathcal{V}$-category. The following are equivalent:*

1. *$x, y \in \underline{\mathcal{C}}$ are isomorphic as objects in $\mathcal{C}$;*
2. *The functors $\mathcal{C}(x, -), \mathcal{C}(y, -) : \mathcal{C} \to \mathcal{S}\mathrm{et}$ are naturally isomorphic;*
3. *The unenriched functors $\underline{\mathcal{C}}(x, -), \underline{\mathcal{C}}(y, -) : \mathcal{C} \to \mathcal{V}$ are naturally isomorphic;*
4. *The $\mathcal{V}$-functors $\underline{\mathcal{C}}(x, -), \underline{\mathcal{C}}(y, -) : \underline{\mathcal{C}} \to \underline{\mathcal{V}}$ are $\mathcal{V}$-isomorphic.*

Proof. By proposition 3.1, underlying is a functor, and hence (4) implies (3). Recall by definition and a previous exercise that $\mathcal{C}(x, -) = \mathcal{V}(*, -) \circ \underline{\mathcal{C}}(x, -)$, and thus (3) implies (2) via horizontal composition of the natural isomorphism with the identity for the underlying functor. By the Yoneda lemma, (2) implies (1). Finally, (1) implies (4) via the following direct construction.

Let $x, y \in \underline{\mathcal{C}}$ be isomorphic in $\mathcal{C}$. Recall a $\mathcal{V}$-natural transformation $\alpha : \underline{\mathcal{C}}(x, -) \Rightarrow \underline{\mathcal{C}}(y, -)$ is a collection of arrows in $\mathcal{V}$ such that for all $a, b \in \underline{\mathcal{C}}$, the diagram

$$\begin{tikzcd}
\underline{\mathcal{C}}(a, b) \arrow[r, "\underline{\mathcal{C}}(x,-)"] \arrow[d, "\underline{\mathcal{C}}(y,-)"'] & \underline{\mathcal{V}}(\underline{\mathcal{C}}(x, a), \underline{\mathcal{C}}(x, b)) \arrow[d, "(\alpha_b)_*"] \\
\underline{\mathcal{V}}(\underline{\mathcal{C}}(y, a), \underline{\mathcal{C}}(y, b)) \arrow[r, "(\alpha_a)^*"'] & \underline{\mathcal{V}}(\underline{\mathcal{C}}(x, a), \underline{\mathcal{C}}(y, b))
\end{tikzcd}$$

commutes. Let $f : y \to x$ be the isomorphism in question, and equivalently $f : * \to \underline{\mathcal{C}}(y, x)$ in $\mathcal{V}$. Since precomposition f^* is an arrow $\underline{\mathcal{C}}(x, -) \to \underline{\mathcal{C}}(y, -)$, define f_a^* as precomposition $\underline{\mathcal{C}}(x, a) \to \underline{\mathcal{C}}(y, a)$. Hence we obtain since $\mathcal{V}$ is closed the adjunction

$$f_a^* \in \mathcal{V}(* \otimes \underline{\mathcal{C}}(x, a), \underline{\mathcal{C}}(y, a)) \cong \mathcal{V}(*, \underline{\mathcal{V}}(\underline{\mathcal{C}}(x, a), \underline{\mathcal{C}}(y, a))).$$

Here we abuse notation and write f^a_* for $f^a_* \circ \lambda_{\underline{\mathcal{C}}(x,a)}$, for λ the left unitor. This remark will be important later. Define α_a as the adjunct of f^*_a above. Let $g = f^{-1}$, and define β_a as the adjunct of g^*_a, forming another transformation $\beta : \underline{\mathcal{C}}(y, -) \Rightarrow \underline{\mathcal{C}}(x, -)$. The claim is that the horizontal composites $\beta \circ \alpha = 1$ and $\alpha \circ \beta = 1$.

Recall the horizontal composite is defined pointwise as

$$(\beta \circ \alpha)_a : * \xrightarrow{\lambda_*} * \otimes * \xrightarrow{\beta_a \otimes \alpha_a} \underline{\mathcal{V}}(\underline{\mathcal{C}}(y,a), \underline{\mathcal{C}}(x,a)) \otimes \underline{\mathcal{V}}(\underline{\mathcal{C}}(x,a), \underline{\mathcal{C}}(y,a)) \xrightarrow{\circ} \underline{\mathcal{V}}(\underline{\mathcal{C}}(x,a), \underline{\mathcal{C}}(x,a)).$$

An annoying expansion of the precompositions shows that this composite is $\mathrm{id}_{\underline{\mathcal{C}}(x,a)}$. It remains to be shown that these defintions are indeed natural, which is fairly trivial. ■

Adjunctions and equivalences can be defined in any 2-category. We use this fact to define them in the 2-categories $\mathcal{V}$-$\mathcal{C}$at, and indeed these lead to the correct notions.

Definition 3.7 ($\mathcal{V}$-equivalence)**.** A $\mathcal{V}$-equivalence of categories is a $\mathcal{V}$-functor $F : \underline{\mathcal{C}} \to \underline{\mathcal{D}}$ that is

- *essentially surjective*: every $d \in \underline{\mathcal{D}}$ is isomorphic in $\mathcal{D}$ to some object $F(c)$;
- $\mathcal{V}$-fully faithful: for each $c, c' \in \underline{\mathcal{C}}$, the arrow $F_{c,c'}$ is an isomorphism in $\mathcal{V}$.

The usual proof shows that this is precisely the same as the other definition; that is a $\mathcal{V}$-equivalence of categories $F : \underline{\mathcal{C}} \to \underline{\mathcal{D}}$ is a $\mathcal{V}$-functor such that there exists a $\mathcal{V}$-functor $G : \underline{\mathcal{D}} \to \underline{\mathcal{C}}$ which is an inverse up to isomorphism:

$$F \circ G = 1_{\underline{\mathcal{D}}}, \qquad G \circ F = 1_{\underline{\mathcal{C}}}.$$

♡

Definition 3.8 ($\mathcal{V}$-adjunction)**.** A $\mathcal{V}$-adjunction consists of $\mathcal{V}$-functors

$$\underline{\mathcal{C}} \underset{G}{\overset{F}{\rightleftarrows}} \underline{\mathcal{D}}$$

together with either of the following equivalent criteria:

- $\mathcal{V}$-natural isomorphisms $\underline{\mathcal{D}}(F(c), d) \cong \underline{\mathcal{C}}(c, G(d))$ in $\mathcal{V}$;
- $\mathcal{V}$-natural transformations $\eta : 1 \Rightarrow GF$ and $\varepsilon : FG \Rightarrow 1$ satisfying the triangle identities

$$G\varepsilon \circ \eta G = 1_G, \qquad \varepsilon F \circ F\eta = 1_F.$$

♡

One example is as follows: we will see later that the localization $\mathrm{s}\mathcal{S}\mathrm{et} \to \mathrm{Ho}(\mathrm{s}\mathcal{S}\mathrm{et} =: \mathcal{H}$ is lax monoidal, and hence there is a functor $\mathrm{s}\mathcal{S}\mathrm{et}$-$\mathcal{C}\mathrm{at} \to \mathcal{H}$-$\mathcal{C}\mathrm{at}$. An equivalence of simplicially enriched categories (called a DK-equivalence for Dwyer and Kan) is a simplicial functor where the corresponding $\mathcal{H}$-functor is a $\mathcal{H}$-equivalence.

3.6 Simplicial Categories

We explore simplicial categories as an example of these notions with important applications to homotopy theory. A simplicial category $\underline{\mathcal{C}}$ induces a simplicial object in $\mathcal{C}$at, denoted $\mathcal{C}_\bullet : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{C}\mathrm{at}$ and defined as follows. The category $\mathcal{C}_n$ takes as objects the same as $\underline{\mathcal{C}}$, and as arrows is defined as

$$\mathcal{C}_n(x, y) \coloneqq \underline{\mathcal{C}}(x, y)_n.$$

Hence, arrows $x \to y$ in $\mathcal{C}_n$ are n-simplicies in the hom-space $\underline{\mathcal{C}}(x, y)$. The induced actions of the morphisms in $\mathbf{\Delta}$ on the hom-spaces define functors which are identities on objects. Also note that the category $\mathcal{C}_0$ is the underlying category of $\underline{\mathcal{C}}$.

For example, recall that sSet is closed cartesian via the definition $\underline{\mathrm{s}\mathcal{S}\mathrm{et}}(X, Y)_n \coloneqq \mathrm{s}\mathcal{S}\mathrm{et}(\Delta^n \times Y, X)$. Hence $\underline{\mathrm{s}\mathcal{S}\mathrm{et}}$ is a simplicial object in (large) categories. Objects of each category are simplicial sets, and morphisms $X \to Y$ in the nth category are morphisms $\underline{\mathrm{s}\mathcal{S}\mathrm{et}}(X, Y)_n$ of simplicial sets with our previous definition. The face and degeneracy operators act by precomposition on the representable part of the domain.

As a fairly obvious partial converse, any simplicial object $\mathcal{C}_\bullet : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{C}\mathrm{at}$ with functors $d_i : \mathcal{C}_n \to \mathcal{C}_{n-1}$, $s_i : \mathcal{C}_n \to \mathcal{C}_{n+1}$ identities on objects corresponds to a simplicial category, with $\underline{\mathcal{C}}(x, y)_n \coloneqq \mathcal{C}_n(x, y)$. The face and degeneracy functors specify the simplicial actions.

Simplicial functors $F : \underline{\mathcal{C}} \to \underline{\mathcal{D}}$ are collections of functors $F_n : \mathcal{C}_n \to \mathcal{D}_n$ that *commute with the simplicial operator functors*, hence having the same underlying object functions. For example, $F : \underline{\mathcal{C}} \to \underline{\mathrm{s}\mathcal{S}\mathrm{et}}$ gives a simplicial set $F(x)$ for each $x \in \underline{\mathcal{C}}$, and a morphism $F_{x,y} : \underline{\mathcal{C}}(x, y) \to \underline{\mathrm{s}\mathcal{S}\mathrm{et}}(x, y)$ for each x, y; equivalently a *function* mapping each n-simplex in $\underline{\mathcal{C}}(x, y)$ to a map $\Delta^n \times F(x) \to F(y)$ natural in n, meaning that face adn degeneracies of the map correspond to the faces and degeneracies of the n-simplex.

A simplicial natural transformation between simplicial functors $F, G : \underline{\mathcal{C}} \to \underline{\mathcal{D}}$ is a collection of arrows in the underlying catergory $\mathcal{D}$ for each object of $\underline{\mathcal{C}}$, satisfying the enriched naturality condition. Here, that enriched naturality condition says that the underlying natural transformation is natural between the underlying functors, but also that the degenerate images form natural transformations. For example, the collection of images $s_0(\alpha_x) \in \mathcal{D}_1(F(x), G(x))$ forms a natural transformation $F_1 \Rightarrow G_1$; and in general the unique degeneracy operator $[n] \to [0]$ forms a natural transformation $F_n \Rightarrow G_n$.

3.7 Tensors and Cotensors

We will now prove a statement that was stated earlier. The proof was delegated to this section for pedagogical purposes. Given $(\mathcal{V}, \times, *)$ closed monoidal[2], we need to show that $\underline{\mathcal{V}}(u \times v, w) \cong \underline{\mathcal{V}}(u, \underline{\mathcal{V}}(v, w))$. Note for any $x \in \mathcal{V}$ that there is a sequence of natural isomorphisms

$$\mathcal{V}(x, \underline{\mathcal{V}}(u \times v, w)) \cong \mathcal{V}(x \times u \times v, w) \cong \mathcal{V}(x \times u, \underline{\mathcal{V}}(v, w)) \cong \mathcal{V}(x, \underline{\mathcal{V}}(u, \underline{\mathcal{V}}(v, w)))$$

where we have used various results discussed previously. Hence, these two elements represent the same functor, and by the Yoneda lemma they are isomorphic. We also have from our previous discussion that for any $v \in \mathcal{V}$, the represented functor $\underline{\mathcal{V}}(v, -) : \mathcal{V} \to \mathcal{V}$ is a $\mathcal{V}$-functor (as viewed as a functor between the enriched structures). It follows that the left adjoint $- \times v : \mathcal{V} \to \mathcal{V}$ also inherits $\mathcal{V}$-functor structure. Given η^v the unit of the adjunction at v, we very simply have

$$\underline{\mathcal{V}}(u, w) \xrightarrow{(\eta^v_w)_*} \underline{\mathcal{V}}(u, \underline{\mathcal{V}}(v, w \times v)) \cong \underline{\mathcal{V}}(u \times v, w \times v).$$

Define the required action on hom-objects as this composition.

This generalizes. Let $\underline{\mathcal{M}}$ and $\underline{\mathcal{N}}$ be $\mathcal{V}$-categories with $F \dashv G$ an adjunction between the underlying categories

$$\mathcal{N}(F(m), n) \cong \mathcal{M}(m, G(n)).$$

[2] Here I adhere to the book's convention of using $\times$ instead of $\otimes$ for the monoidal product. This means $\times$ does not denote the product, but a monoidal product.

We can try to enrich this adjunction to an isomorphism in $\mathcal{V}$. If we want to use the same idea as previously, meaning we want $\underline{\mathcal{N}}(F(m), n)$ and $\underline{\mathcal{M}}(m, G(n))$ to represent the same functor, leads us to consider $\mathcal{V}(v, \underline{\mathcal{M}}(m, G(n)))$. However there's not a clear way to proceed from here, and we require some special adjunction conditions. We can prove this if each $\underline{\mathcal{M}}(m, -)$ and $\underline{\mathcal{N}}(n, -)$ has a left adjoint preserved by F, or dually if the contravariant hom-functors have right adjoints preserved by G. We will return to this situation shortly after working through the definitions this motivates.

Definition 3.9 (Tensored)**.** A $\mathcal{V}$-category $\underline{\mathcal{M}}$ is *tensored* if for each $v \in \mathcal{V}$ and $m \in \mathcal{M}$, there is an object $v \otimes m \in \mathcal{M}$ and isomorphisms

$$\underline{\mathcal{M}}(v \otimes m, n) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{M}}(m, n)).$$

♡

By Yoneda, there is a unique way to make $\otimes : \mathcal{V} \times \mathcal{M} \to \mathcal{M}$ a bifunctor natural in all three variables. Hence each $\mathcal{M}(m, -)$ admits a left $\mathcal{V}$-adjoint $- \otimes m$.

Definition 3.10 (Cotensored)**.** A $\mathcal{V}$-category $\underline{\mathcal{M}}$ is *cotensored* if for each $v \in \mathcal{V}$ and $m \in \mathcal{M}$, there is an object n^v and isomorphisms

$$\underline{\mathcal{M}}(m, n^v) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{M}}(m, n)).$$

♡

The dual statements are true: there is a unique bifunctor $(-)^{-} : \mathcal{V}^{\mathrm{op}} \times \mathcal{M} \to \mathcal{M}$ with isomorphisms natural in all variables, and each $\underline{\mathcal{M}}(-, n)$ has a right $\mathcal{V}$-adjoint n^{-}.

For example, a closed $\mathcal{V}$-category is a monoidal category which is tensored, with $\otimes = \times$ simply the monoidal product, and which is cotensored with $\underline{\mathcal{V}}(u, v) = v^u$. Hence any locally small category with products and coproducts is enriched, tensored, and cotensored over $\mathcal{S}et$ with the usual cartesian monoidal structure: the cotensore is the copower, and the tensor is the power.

Lemma 3.5. *Let $(\mathcal{V}, \times, *)$ be closed symmetric monoidal, and let $\mathcal{M}$ be a tensored $\mathcal{V}$-category. Then tensor product is unital and associative.*

Proof. Recall that we have natural isomorphisms

$$\mathcal{V}(w, \underline{\mathcal{V}}(*, v)) \cong \mathcal{V}(w \times *, v) \cong \mathcal{V}(w, v).$$

Thus $\mathcal{V}(*, v)$ and v represent the same functor, and are thus isomorphic by the Yoneda lemma. Thus we obtain

$$\underline{\mathcal{M}}(* \otimes m, n) \cong \underline{\mathcal{V}}(*, \underline{\mathcal{M}}(m, n)) \cong \underline{\mathcal{M}}(m, n).$$

Therefore, $* \otimes m$ and m represent the same $\mathcal{V}$-functor, so by lemma 3.4 we have $* \otimes m \cong m$. Sincew these isomorphisms are natural, we have that there is a natural isomorphism $* \otimes - \cong 1$, hence $\otimes$ is unital.

Similarly we obtain

$$\mathcal{M}((v \times w) \otimes m, n) \cong \mathcal{V}(v \times w, \underline{\mathcal{M}}(m, n)) \cong \mathcal{V}(v, \underline{\mathcal{V}}(w, \underline{\mathcal{M}}(m, n))) \cong \mathcal{V}(v, \underline{\mathcal{M}}(w \otimes m, n)) \cong \mathcal{M}(v \otimes (w \otimes m), n).$$

This proves the associativity property

$$(v \times w) \otimes m \cong v \otimes (w \otimes m).$$

Proposition 3.2. *Let $\underline{\mathcal{M}}, \underline{\mathcal{N}}$ be tensored and cotensored $\mathcal{V}$-categories and $F : \mathcal{M} \rightleftarrows \mathcal{N} : G$ an adjunction between the underlying categories. Then any of the following determines the others:*

- *A $\mathcal{V}$-adjunction $\underline{\mathcal{N}}(F(m), n) \cong \underline{\mathcal{M}}(m, G(n))$;*
- *A $\mathcal{V}$-functor F and natural isomorphisms $F(v \otimes m) \cong v \otimes F(m)$;*
- *A $\mathcal{V}$-functor G and natural isomorphisms $G(n^v) \cong G(n)^v$.*

Proof. Left as an exercise. It suffices to show they are equivalent criteria. First we show 1 implies 2. Let $F \dashv G$ induce a $\mathcal{V}$-adjunction. We want to show $F(v \otimes m) \cong v \otimes F(m)$. By the $\mathcal{V}$-adjunction, we have

$$\underline{\mathcal{N}}(F(v \otimes m), n) \cong \mathcal{M}(v \otimes m, G(n)).$$

By definition of a tensored category, we have

$$\underline{\mathcal{M}}(v \otimes m, G(n)) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{M}}(m, G(n))).$$

By lemma 3.4, since $\underline{\mathcal{M}}(m, G(n)) \cong \underline{\mathcal{N}}(F(m), n)$ are isomorphic, they represent the same $\mathcal{V}$-functor, hence

$$\underline{\mathcal{V}}(v, \underline{\mathcal{M}}(m, G(n))) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{N}}(F(m), n)) \cong \underline{\mathcal{V}}(v \otimes F(m), n),$$

where the last isomorphism once again follows by definition of the tensor. Thus, $F(v \otimes m)$ and $v \otimes F(m)$ represent the same $\mathcal{V}$-functor, so by lemma 3.4 they are isomorphic. Sinec all the isomorphisms here were natural, so is that isomorphism.

Now we show 2 implies 3. Since $\mathcal{M}$ is tensored and cotensored,

$$\underline{\mathcal{M}}(v \otimes m, n) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{M}}(m, n)) \cong \underline{\mathcal{M}}(m, n^v).$$

The same is true for $\mathcal{N}$. The following sequence of isomorphisms makes heavy use of this fact, and makes use of 2-functorality of the underlying object functor to lower these $\mathcal{V}$-adjunctions to regular adjunctions. We have

$$\begin{aligned}
\mathcal{N}(F(v \otimes m), n) &\cong \mathcal{N}(v \otimes F(m), n) \\
&\cong \mathcal{V}(v, \underline{\mathcal{N}}(F(m), n)) \\
&\cong \mathcal{V}(v, \underline{\mathcal{M}}(m, G(n)) \\
&\cong \mathcal{M}(m, G(n)^v) \\
\mathcal{V}(v, \underline{\mathcal{N}}(F(m), n)) &\cong \mathcal{N}(F(m), n^v) \\
&\cong \mathcal{M}(m, G(n^v)).
\end{aligned}$$

By the same logic as before, we have the isomorphism, as required.

Now we show 3 implies 1. Since $G(n^v) \cong G(n)^v$, we have

$$\mathcal{M}(m, G(n^v)) \cong \mathcal{M}(m, G(n)^v) \cong \mathcal{V}(v, \underline{\mathcal{M}}(m, G(n))).$$

We also have since $F \dashv G$

$$\mathcal{M}(m, G(n^v)) \cong \mathcal{N}(F(m), n^v) \cong \mathcal{V}(v, \underline{\mathcal{N}}(F(m), n)).$$

By the same logic as previously, we are done. ∎

Theorem 3.1. *Let $F : \mathcal{V} \rightleftarrows \mathcal{U} : G$ be an adjunction between closed symmetric monoidal categories such that F is strong monoidal. Then any tensored, cotensored, and enriched $\mathcal{U}$-category can be canonically enriched, tensored, and cotensored over $\mathcal{V}$.*

Proof. Some paper from 1974 shows that F strong monoidal implies G is lax monoidal, and hence by the change of base theorem, or more generally proposition 3.1, we can change the base of enrichment from $\mathcal{U}$ to $\mathcal{V}$. Given that $\mathcal{M}$ is a tensored and cotensored $\mathcal{U}$-category

$$\underline{\mathcal{M}}_{\mathcal{U}}(u \otimes m, n) \cong \underline{\mathcal{U}}(u, \underline{\mathcal{M}}_{\mathcal{U}}(m, n)) \cong \underline{\mathcal{M}}_{\mathcal{U}}(m, n^u),$$

we define the tensor $- \star - : \mathcal{V} \times \mathcal{M} \to \mathcal{M}$, cotensor $- \pitchfork - : \mathcal{V}^{\mathrm{op}} \times \mathcal{M} \to \mathcal{M}$, and internal hom $\underline{\mathcal{M}}_{\mathcal{V}}(-, -) : \mathcal{M}^{\mathrm{op}} \times \mathcal{M} \to \mathcal{V}$ by

$$v \star m := F(v) \otimes m, \qquad v \pitchfork m := m^{F(v)}, \qquad \underline{\mathcal{M}}_{\mathcal{V}}(m, n) := G\underline{\mathcal{M}}_{\mathcal{U}}(m, n).$$

We will show the required natural isomorphism

$$\underline{\mathcal{M}}_{\mathcal{V}}(v \star m, n) \cong \underline{\mathcal{V}}(v, \underline{\mathcal{M}}_{\mathcal{V}}(m, n)) \cong \underline{\mathcal{M}}(m, v \pitchfork n)$$

in the same way as before, by showing they represent the same functor $\mathcal{V}^{\mathrm{op}} \to \mathcal{S}\mathrm{et}$. The first isomorphism is given as

$$\begin{aligned}
\mathcal{V}(x, \underline{\mathcal{M}}_{\mathcal{V}}(v \star m, n)) &= \mathcal{V}(x, G\underline{\mathcal{M}}_{\mathcal{U}}(F(v) \otimes m, n)) \\
&\cong \mathcal{U}(F(x), \underline{\mathcal{M}}_{\mathcal{U}}(F(v) \otimes m, n)) \\
&\cong \mathcal{M}(F(x) \otimes (F(v) \otimes m), n) \\
&\cong \mathcal{M}((F(x) \times F(v)) \otimes m, n) \\
&\cong \mathcal{U}(F(x) \times F(v), \underline{\mathcal{M}}_{\mathcal{U}}(m, n)) \\
&\cong \mathcal{U}(F(x \times v), \underline{\mathcal{M}}_{\mathcal{U}}(m, n)) \\
&\cong \mathcal{V}(x \times v, G\underline{\mathcal{M}}_{\mathcal{U}}(m, n)) \\
&= \mathcal{V}(x \times v, \underline{\mathcal{M}}_{\mathcal{V}}(m, n)) \\
&\cong \mathcal{V}(x, \underline{\mathcal{V}}(v, \underline{\mathcal{M}}_{\mathcal{V}}(m, n))).
\end{aligned}$$

The second isomorphism is given by going the reverse direction. Note that dualizing associativity of the tensor gives the property

$$n^{u \times v} \cong (n^u)^v.$$

Hence we obtain

$$\begin{aligned}
\mathcal{V}(x, \underline{\mathcal{V}}(v, \underline{\mathcal{M}}_{\mathcal{V}}(m,n))) &\cong \mathcal{V}(x \times v, \underline{\mathcal{M}}_{\mathcal{V}}(m,n)) \\
&= \mathcal{V}(x \times v, G\underline{\mathcal{M}}_{\mathcal{U}}(m,n)) \\
&\cong \mathcal{U}(F(x \times v), \underline{\mathcal{M}}_{\mathcal{U}}(m,n)) \\
&\cong \mathcal{U}(F(x) \times F(v), \underline{\mathcal{M}}_{\mathcal{U}}(m,n)) \\
&\cong \mathcal{M}(m, n^{F(x)\times F(v)}) \\
&\cong \mathcal{M}(m, n^{F(v)\times F(x)}) \\
&\cong \mathcal{M}(m, (n^{F(v)})^{F(x)}) \\
&= \mathcal{M}(m, (v \pitchfork n)^{F(x)}) \\
&\cong \mathcal{U}(F(x), \underline{\mathcal{M}}_{\mathcal{U}}(m, v \pitchfork n)) \\
&\cong \mathcal{V}(x, G\underline{\mathcal{M}}_{\mathcal{U}}(m, v \pitchfork n)) \\
&= \mathcal{V}(x, \underline{\mathcal{M}}_{\mathcal{V}}(m, v \pitchfork n)).
\end{aligned}$$

■

Corollary 3.1. *The adjunction $F \dashv G$ of the previous* **??**, *hereafter called a* strong monoidal adjunction, *is a $\mathcal{V}$-adjunction with respect to the $\mathcal{V}$-structure induced on $\mathcal{U}$.*

Proof. By the yoneda lemma and the definitions above,

$$\begin{aligned}
\mathcal{V}(w, \underline{\mathcal{U}}_{\mathcal{V}}(F(v), u)) \cong \mathcal{V}(w, G\underline{\mathcal{U}}_{\mathcal{U}}(F(v), u)) \cong \mathcal{U}(F(w), \underline{\mathcal{U}}_{\mathcal{U}}(F(v), u) \cong \mathcal{U}(F(w) \times F(v), u) \\
\cong \mathcal{U}(F(w \times v), u) \cong \mathcal{V}(w \times v, G(u)) \cong \mathcal{V}(w, \underline{\mathcal{V}}(v, G(u))).
\end{aligned}$$

■

Results following from this show that the categories of based simplicial sets and based topological spaces are enriched, tensored, and cotensored. For example, the cotensor Y^K in $\underline{\mathrm{s}\mathcal{S}\mathrm{et}}_*$ is $\underline{\mathrm{s}\mathcal{S}\mathrm{et}}_*(K_+, Y)$.

More interestingly, we will show later that the geometric realization is a strong monoidal left adjoint $|-| \dashv S$. Because the convenient category of topological spaces $\mathcal{T}\mathrm{op}$ is closed monoidal, it is thus tensored, cotensored, and enriched over simplicial sets with definitions as $K \otimes X \coloneqq |K| \times X$ and $X^K \coloneqq \underline{\mathcal{T}\mathrm{op}}(|K|, X)$. The simplicial enrichment has as hom-objects $S\left(\underline{\mathcal{T}\mathrm{op}}(X,Y)\right)$. Hence this is a simplicial adjunction.

Exercise. Let $\mathcal{M}$ be cocomplete. Show that $\mathcal{M}^{\mathbf{\Delta}^{\mathrm{op}}}$, the category of simplicial objects in $\mathcal{M}$, is simplicially enriched and tensored, with $(K \otimes X)_n \coloneqq K_n \cdot X_n$ the copower.

Proof. The proof is messy and I don't want to copy it down. The internal hom is defined the same way as the closed structure on $\mathrm{s}\mathcal{S}\mathrm{et}$, as one would expect:

$$\underline{\mathcal{M}^{\mathbf{\Delta}^{\mathrm{op}}}}(X,Y)_n \coloneqq \mathcal{M}^{\mathbf{\Delta}^{\mathrm{op}}}(\Delta^n \otimes X, Y).$$

Simplicial maps are induced in the expected way via precomposition. Since a closed structure is the monoidal product also being a tensor, this definition agrees with the definition for the closed monoidal structure on $\mathrm{s}\mathcal{S}\mathrm{et}$. ■

For example, an unenriched adjunction $F : \mathcal{M} \rightleftarrows \mathcal{N} : G$ promotes to an adjunction given by levelwise postcomposition

$$F_* : \mathcal{M}^{\mathbf{\Delta}^{\mathrm{op}}} \rightleftarrows \mathcal{N}^{\mathbf{\Delta}^{\mathrm{op}}} : G_*.$$

Since F_* is left adjoint, it preserves coproducts, and thus tensors. By proposition 3.2, $F_* \dashv G_*$ is thus a simplicial adjunction.

Another interesting result is the Dold-Kan correspondence: there is an equivalence of categories $\mathcal{A}\mathrm{b}^{\mathbf{\Delta}^{\mathrm{op}}} \rightleftarrows \mathcal{C}\mathrm{h}_{\geq 0}(\mathbb{Z})$.

3.8 Simplicial homotopy and simplicial model categories

If $\underline{\mathcal{M}}$ is a simplicial category, we obtain a canonical notion of homotopy in the underlying category $\mathcal{M}$. Recall that Δ^0 is the unit for the monoidal product in $s\mathcal{S}\mathrm{et}$. A *homotopy* between morphisms $f, g : m \to n$ is given as the commutative diagram

$$\begin{tikzcd}
\Delta^0 \arrow[d, "d^1"'] \arrow[dr, "f"] & \\
\Delta^1 \arrow[r, "H"] & \underline{\mathcal{M}}(m,n) \\
\Delta^0 \arrow[u, "d^0"] \arrow[ur, "g"'] &
\end{tikzcd}$$

If $\underline{\mathcal{M}}$ is tensored, then we can realize the homotopy as a homotopy in the underlying category by recalling that we have an isomorphism

$$s\mathcal{S}\mathrm{et}(-, \underline{\mathcal{M}}(m,n)) \cong \mathcal{M}(- \otimes m, n).$$

Thus the diagram transposes as

$$\begin{tikzcd}
\Delta^0 \otimes m \cong m \arrow[d, "d^1 \otimes 1"'] \arrow[dr, "f"] & \\
\Delta^1 \otimes m \arrow[r, "H"] & n \\
\Delta^0 \otimes m \cong m \arrow[u, "d^0 \otimes 1"] \arrow[ur, "g"'] &
\end{tikzcd}$$

Similarly, if $\mathcal{M}$ is cotensored, then we obtain a homotopy in $\mathcal{M}$ as

$$\begin{tikzcd}
& n \cong n^{\Delta^0} \\
m \arrow[ur, "f"] \arrow[r, "H"] \arrow[dr, "g"'] & n^{\Delta^1} \arrow[u, "n^{d^1}"'] \arrow[d, "n^{d^0}"] \\
& n \cong n^{\Delta^0}
\end{tikzcd}$$

Thus we obtain a notion of a simplicial homotopy equivalence if $\mathcal{M}$ is either tensored or cotensored over simplicial sets.

Definition 3.11 (Geometric Realization)**.** If $\mathcal{M}$ is simplicially enriched, tensored, and cocomplete, the *geometric realization* of a simplicial object $X_\bullet$ in $\mathcal{M}$ is

$$|X_\bullet| := \int^{n\in\mathbf{\Delta}} \Delta^n \otimes X_n.$$

These coends define a functor $|-| : \mathcal{M}^{\mathbf{\Delta}^{\mathrm{op}}} \to \mathcal{M}$. ♡

If $\mathcal{M}$ is simplicially tensored, there are two reasonable definitions for simplicial tensors on $\mathcal{M}^{\mathbf{\Delta}^{\mathrm{op}}}$. One can define tensors pointwise for any diagram category $\mathcal{M}^{\mathcal{D}}$: given $K \in \mathrm{s}\mathcal{S}\mathrm{et}$ and $X : \mathcal{D} \to \mathcal{M}$, define $K \otimes X$ as the functor $d \mapsto K \otimes X(d)$. The category of simplicial objects is then a special case thereof. However, we prefer to define tensores as done earlier, via $(K \otimes X)_n := K_n \cdot X_n$.

Lemma 3.6. *Let $\mathcal{M}$ be simplicially enriched, tensored, cotensored, and admits geometric realizations. Then geometric realization preserves tensors.*

Will be proven later. The key is that tensor admits pointwise right adjoints, and thus preserves colimits in both variables. We present this result now due to the following corollary.

Corollary 3.2. *If $\mathcal{M}$ is simplicially enriched, tensored, and cotensored, then geometric realization preserves simplicial homotopy equivalences.*

> **Proof.** Given a simplicial homotopy equivalence $fg, gf \sim 1$ between simplicial objects $X_\bullet, Y_\bullet$ in $\mathcal{M}^{\mathbf{\Delta}^{\mathrm{op}}}$, since geometric realization preserves tensors, geometric realization produces simplicial homotopy equivalences in $\mathcal{M}$ between the geometric realizations of the simplicial objects. ■

The chapter is ended with the motivation we sought at the end of the previous chapter. The following lemma defines some properties of simplicial model categories, which will be properly introduced later. This result suffices as a working understanding for now.

Lemma 3.7. *A simplicial model category*

- *is complete and cocomplete;*
- *is tensored, cotensored, and enriched over simplicial sets;*
- *has subcategories of* cofibrant *and* fibrant *objects preserved by tensoring and cotensoring, respectively, with any simplicial set;*
- *admits a left deformation Q into the cofibrant objects and a right deformation R into the fibrant objects;*
- *is a saturated homotopical category;*
- *has simplicial homotopy equivalences among the weak equivalences;*
- *has the property that the internal hom preserves weak equivalences between cofibrant objects in its first variable provided the second variable is fibrant, and preserves weak equivalences between fibrant objects in its second variable provided the first is cofibrant.*

Examples include $\mathrm{s}\mathcal{S}\mathrm{et}, \mathrm{s}\mathcal{S}\mathrm{et}_*, \mathcal{T}\mathrm{op}, \mathcal{T}\mathrm{op}_*$. There are analogous definitions of $\mathcal{V}$-model categopries, which are tensored, cotensored, and enriched over $\mathcal{V}$ instead of $\mathrm{s}\mathcal{S}\mathrm{et}$. A homotopical version of theorem 3.1 says that if the strong monoidal adjunction is Quillen, then any $\mathcal{U}$-model category is a $\mathcal{V}$-model category. The adjunctions

$$(-)_+ : \mathrm{s}\mathcal{S}\mathrm{et} \rightleftarrows \mathrm{s}\mathcal{S}\mathrm{et}_* : U, \qquad |-| : \mathrm{s}\mathcal{S}\mathrm{et} \rightleftarrows \mathcal{T}\mathrm{op} : S$$

are strong monoidal and Quillen, provided $\mathcal{T}op$ is once again a convenient category. Hence, any based simplicial or topological model category is a simplicial model cateogry. The converse is not always true however. We determine that simplicial model categories are an appropriate place to define homotopy limits and colimits.